STUDIO EMPIRE

Full Campaign Progression Bible

Implementation authority for the complete journey from founder garage to an eight-seat future technology campus

LOCK	AUTHORITATIVE RULE
Progression	1 -> 4 -> 5 -> 6 -> 8 total HQ seats
Presentation	One full-size focus character; all other staff may work remotely
Final expansion	12 Technology Park years; no graphics/GPU gate
Leadership	R&D; Director and Hardware Director occupy the final two role-locked seats
Campaigns	35-year Classic and 50-year Legacy through the 2050 future era
Postgame	True Endless Mode with continuing markets, platforms, staff, and technology

18,000+ words | 40 sections | 150 acceptance tests | Version 1.0 | August 7, 2026

Contents

Sections and subsections are bookmarked for fast navigation on mobile PDF viewers.

0. How to use this document

This document defines how Studio Empire progresses from a one-person garage to an eight-seat technology campus, how each transition is earned, what each tier unlocks, how employees and remote teams work on a vertical screen, how R&D and Hardware become real managed divisions, how AAA and online games evolve into subscription and persistent-world businesses, and how the game continues after its scored campaign.

Treat this document as authoritative unless a newer signed-off specification explicitly replaces a rule. Numeric values marked initial tuning value belong in configuration and may be balanced without changing the dependency structure. Do not invent a second progression path in UI code.

The current Garage Phase remains valid and must not be rebuilt. This specification begins at the same canonical state and extends it.

Existing rules that remain binding

1. The simulation is domain-first, deterministic, seedable, saveable, and replayable.
2. UI components consume read-only view models and dispatch commands. They never calculate reviews, sales, money, work, research, progression, or unlocks.
3. The founder is a distinct entity, not a hired employee with a special name.
4. The founder has no employee-energy meter.
5. Hired employees may have energy, fatigue, morale, burnout risk, salary, training, specialization, and assignments.
6. The six main genres remain Action, Adventure, RPG, Simulation, Strategy, and Casual.
7. Production remains three stages with nine disciplines:
   - Stage 1: Engine, Gameplay, Story & Quests
   - Stage 2: Dialogue, Level Design, AI
   - Stage 3: World Design, Graphics, Sound
8. Each discipline uses an independent 0-100 intent value. The values are not forced to sum to 100.
9. Higher intent creates real work, time, cost, bugs, and staffing demand. It does not directly type quality into the score.
10. The commercial chain remains separated and explainable:

```
Concept Fit -> Production Demand -> Execution Quality -> Reviews -> Market Demand -> Weekly Sales
-> Fans/Knowledge
```

11. Marketing changes awareness, reach, and hype. It never changes execution quality or review truth.
12. Research discovery, research completion, engine integration, engine versions, game projects, platform builds, release, reviews, sales, and post-release support remain separate states.
13. Released games preserve the exact engine version, platform build, features, assignments, and decisions that created them.
14. Office movement preserves the complete campaign. It does not grant free employees, free expertise, free research, or free quality.
15. Distant systems remain invisible until their complete core loop is usable.

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1. Verified Game Dev Tycoon baseline and intentional deviations

The original game is a design reference, not a template to copy blindly. Its strongest progression idea is that the studio itself evolves as the player proves mastery. Its weakest late-game behavior is that the scored timeline ends while the continuing market largely stops evolving.

1.1 What the original actually does

Greenheart's official description says the player starts in an 80s garage, develops simple games, gains experience, creates an engine, moves into an office after successful releases, hires and trains staff, develops larger games, and eventually unlocks labs for major projects. The official FAQ also confirms that the player may continue after the scored campaign, but no more normal platform releases occur. See Greenheart Games' official overview and FAQ.

The commonly cited office progression is:

Original tier	Verified baseline
Garage	Founder alone. First office opportunity appears after reaching roughly $1M.
First office	Move costs roughly $150K. The original can hire up to four employees in addition to the founder.
In-place technology upgrade	Same office, upgraded computers. Requires time in the office, staff, and cash.
Large office / technology park	Requires four staff, roughly $16M liquid cash, and an $8M move cost in the commonly documented rules. Maximum normal team is founder plus six employees.
R&D Lab	Requires the large office and a Design Specialist.
Hardware Lab	Requires the R&D Lab, Hardware research, and a Technology Specialist.

The detailed office thresholds are documented in this Steam office-requirements guide. The original's basic employee stats, training tiers, specialist requirements, and discipline-specialization thresholds are cataloged in this GameFAQs training guide.

The original AAA path is also layered rather than automatic: the player needs an R&D Lab and must first prove a near-perfect Large game before AAA becomes an R&D project. See the documented AAA unlock path.

Original campaign length settings scale platform and research timing. When the timer ends, the score is calculated and play may continue. See the Greenheart forum explanation of campaign-length behavior.

1.2 Deliberate Studio Empire changes

Area	Original reference	Studio Empire lock
First-office capacity	Founder + up to 4 employees	4 HQ seats total: founder + 3 hires
Upgraded first office	Same office/computer upgrade	Same physical office, real Training Center, 5 HQ seats total
First technology park	Founder + up to 6 hires	6 HQ seats total
Final campus	No later office tier	Same technology park expanded after 12 years, 8 HQ seats total
Final two seats	Normal staff	Role-locked R&D Director and Hardware Director seats
Main scene	Several characters visible	Exactly one full-size focus character visible; everyone else may work remotely
Office gate	Often heavily money/time driven	Offer requires money, releases, capability, and runway; date alone never grants mastery
Technology Park expansion gate	Not present	Twelve years of campus tenure plus affordability; no Level-7 graphics-card gate
Research	RP purchase often immediately useful	Discovery -> research -> prototype/integration -> versioned capability
AAA	Larger size	A managed production class with preproduction, milestones, QA, launch, and support
MMO	Larger project with maintenance curve	A persistent-service business with capacity, retention, churn, subscriptions, content, community, and shutdown decisions
Own hardware	Custom-console feature	A full platform lifecycle: architecture, cost, manufacturing, launch lineup, installed base, ecosystem, revisions, and retirement

Area	Original reference	Studio Empire lock
Post-score play	Continue in a mostly frozen market	True endless simulation with procedural platforms, technologies, trends, rivals, staff actions, and market movement

The capacity numbers above are total HQ seats, including the founder. This wording is mandatory everywhere. Never mix "employees," "staff," and "seats" without stating whether the founder and directors are included.

2. The complete five-tier studio ladder

```
Tier 1  Garage                 1 HQ seat
Tier 2  First Office          4 HQ seats
Tier 3  Upgraded Office       5 HQ seats
Tier 4  Technology Park       6 HQ seats + starter remote lab capacity
Tier 5  Expanded Tech Campus  8 HQ seats + expanded remote lab capacity
```

2.1 Capacity definitions

Tier	HQ seats total	HQ composition at full staffing	Remote lab slots	Normal production ceiling
1. Garage	1	Founder	0	1 person
2. First Office	4	Founder + 3 production employees	0	4 people
3. Upgraded Office	5	Founder + 4 production employees	0	5 people
4. Technology Park	6	Founder + 5 production employees	2 R&D + 2 Hardware after each lab is built	6 people, reduced when an HQ specialist acts as a lab lead
5. Expanded Tech Campus	8	Founder + 5 production employees + R&D Director + Hardware Director	4 R&D + 4 Hardware	6-person production group remains intact

Remote lab slots are real staff records with salaries, skills, energy, training, assignments, and departure behavior. They do not appear as full-size characters in the main vertical scene and do not generate normal production-team weekly RP. Their work belongs to their lab.

2.2 What an office tier controls

An office is a capability and capacity container. It controls:

- HQ seat capacity.
- Remote lab capacity.
- Which staff roles may be hired or promoted.
- Normal and auxiliary project concurrency.
- Equipment and engine-integration capacity.
- Available training tiers.
- Available game sizes and publishing options.
- Whether R&D, Hardware, AAA, persistent services, and platform projects may exist.
- Fixed operating overhead.
- The physical background and contextual destinations in the vertical UI.

It does not directly improve game quality. Better facilities create opportunities; people and decisions create results.

3. Campaign length, historical eras, and the 1980-2050 problem

The phrase "1980 to 2050" describes 70 calendar years. A 35-year and a 50-year campaign therefore cannot both use a literal one-simulation-year-equals-one-calendar-year clock and still reach 2050. Do not hide that contradiction inside event data.

Studio Empire uses two clocks:

1. Campaign time is the actual simulation clock: days, weeks, months, and Campaign Year.
2. Industry era is the authored historical/future sequence used for platforms, technology, trends, and fiction.

3.1 Player-facing modes

Mode	Scored campaign	Authored industry span before score	Purpose
Classic	35 campaign years	1980 through approximately 2030	Faster, tighter run; garage through AAA/basic live services; future content remains available after score in Endless Mode
Legacy	50 campaign years	1980 through 2050	Full intended arc; extra time for the 12-year campus expansion, directors, advanced hardware, subscriptions, and future-era projects before scoring

The UI may display `Y12 M4 W2 • Industry Era: late 1990s`. It must not pretend Campaign Year 12 literally equals 1992.

3.2 Timeline scaling

```
type CampaignMode = "classic_35" | "legacy_50"

const campaignYears = mode === "classic_35" ? 35 : 50
const industryEndYear = mode === "classic_35" ? 2030 : 2050

campaignProgress = elapsedCampaignWeeks / totalCampaignWeeks
industryYear = round(1980 + campaignProgress * (industryEndYear - 1980))
```

Authored events use normalized progress or a reference 50-year week. Tutorial and first-office pacing use bounded opening windows and are not stretched into boredom in the Legacy campaign.

3.3 Major expected windows

These are pacing targets, not automatic grants:

Milestone	Classic expected window	Legacy expected window	Hard dependency
First-office offer	Y3-Y6	Y3-Y6	Complete garage proof + cash/runway
Upgraded-office offer	Y8-Y14	Y9-Y16	Staffed office + team/cash proof
Technology Park offer	Y13-Y21	Y16-Y25	Full Tier-3 operation + scale proof
R&D Lab	Y15-Y24	Y18-Y29	Technology Park + Design Specialist
Hardware Lab	Y17-Y27	Y20-Y33	R&D + Hardware research + Technology Specialist
AAA path	Y18-Y30	Y21-Y36	Large-game proof + R&D capability
Basic MMO/live-service path	Y21-Y34	Y24-Y40	Network research + operations capability
Expanded Campus offer	12 years after Technology Park	12 years after Technology Park	Campus tenure + affordability only

Milestone	Classic expected window	Legacy expected window	Hard dependency
Advanced subscriptions/future platforms	Usually Endless	Y35-Y50	Tier 5 + director-led research

If a player reaches a milestone earlier through excellent play, the eligibility window may open at its earliest bound. If a player reaches the date without capability, nothing is granted. If a player reaches the campaign score without Tier 5, Endless Mode keeps the path available.

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4. Universal progression-gate model

Every major transition has five separate states:

```
type UnlockState =
  | "hidden"
  | "discovered"
  | "eligible"
  | "offered"
  | "accepted"
  | "completed"
```

1. Hidden: the player cannot see or reach it.
2. Discovered: news, a mentor, an employee, or studio performance introduces the possibility.
3. Eligible: every non-financial capability requirement is met.
4. Offered: the concrete cost, overhead, runway impact, and benefits are presented.
5. Accepted: money is reserved, construction/move work starts, and cancellation rules apply.
6. Completed: capacity and capabilities change atomically.

4.1 Offer behavior

- Declining an office offer never destroys it.
- The offer remains available from the studio destination and may be reconsidered later.
- Reopening the offer does not reroll price or requirements.
- An offer cannot interrupt release, bankruptcy settlement, or another atomic transaction.
- The player may remain indefinitely in an older office and accept its limitations.
- The move screen must show current versus new capacity, cost, ongoing overhead, required construction time, cash remaining, and estimated runway.
- No capability becomes usable before the move/renovation transaction completes.
- Employees, active assignments, projects, games, engines, labs, and market history survive the transition.

4.2 Affordability and runway

Do not define "can afford" as `cash >= purchaseCost`.

```
cashAfterMove = cash - moveCost - committedSigningBonuses
weeklyBurnAfterMove = newOfficeOverhead + payroll + debtService + activeLabBudgets
runwayWeeks = cashAfterMove / max(1, weeklyBurnAfterMove - trailingWeeklyIncome)
```

An offer may be visible while runway is unsafe, but acceptance requires the configured minimum unless the player explicitly enables a high-risk difficulty rule. Default minimum runway:

- Tier 2: 26 weeks.
- Tier 3: 39 weeks.
- Tier 4: 52 weeks.
- Tier 5: 104 weeks.

4.3 Default progression values

These are initial tuning values, not component-level constants:

Transition	Non-financial proof	Liquid-cash gate	Cost	Minimum tenure
Garage -> First Office	5 releases, 1,000 fans, one profitable title, positive trailing 13-week operating cash flow	$1,000,000	$150,000	Earliest Y3
First Office -> Upgraded Office	All 4 HQ seats filled, 4 releases since move, team cohesion >=70%, Management Fundamentals completed	$5,000,000	$500,000	52 weeks in Tier 2
Upgraded Office -> Technology Park	All 5 HQ seats filled, 12 total releases, 3 profitable Medium-or-larger games, one trained discipline specialist, positive trailing 52-week operating profit	$16,000,000	$8,000,000	104 weeks in Tier 3
Technology Park -> Expanded Campus	No active insolvency, no unpaid lab debt, construction can be funded with 104 weeks of post-expansion runway	$100,000,000	$50,000,000	12 full campaign years in Tier 4

The Tier-5 offer intentionally has no review-score, graphics-version, GPU-level, or lab-project requirement. The user explicitly discarded the Level-7 graphics-card idea. Twelve years creates the opportunity; cash and runway determine whether the company can take it.

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5. Tier 1: Founder Garage

5.1 Entry state

- One founder.
- One HQ seat.
- One active primary assignment.
- Small games only.
- Starter toolkit/imported engine.
- Six main genres.
- Four seed-selected starting topics.
- Early platform market.
- Basic contracts, research discovery, game reports, engine creation, finances, history, save, and settings as currently implemented or scheduled in the Garage specification.
- No visible employees, staff room, office management, R&D, Hardware, AAA, MMO, or future-campus screens.

5.2 Core player loop

1. Select topic, genre, platform, engine, and small scope.
2. Plan and execute the three production stages.
3. Polish and resolve bugs.
4. Enter Pre-Release.
5. Release or cancel.
6. Reveal reviews only after release.
7. Settle sales only after at least seven market days.

8. Decide whether to develop another game, perform a report, research, create an engine, or take a contract.
9. Accumulate money, fans, knowledge, mastery, and RP.
10. Reach and accept the first-office opportunity.

5.3 Garage constraints

- The founder cannot develop and research simultaneously.
- A Game Report consumes the founder's assignment and calendar time.
- Contract work consumes the founder's assignment.
- Released games continue selling while the founder works on something else.
- The founder has no energy bar, but excessive scope creates more work and time.
- The garage cannot hire staff through a hidden menu or direct URL.
- The office offer is a goal, not a forced tutorial completion.

5.4 First-office event flow

When the gate becomes eligible:

1. Finish the current atomic action.
2. Queue one high-priority "A real office is now possible" event.
3. Show the offer summary, not a mandatory move.
4. Let the player inspect:
   - 4 total HQ seats.
   - Hiring and payroll.
   - Basic employee training.
   - Medium-game path.
   - Higher fixed overhead.
   - Move cost and runway.
5. Accept, decline, or decide later.
6. If accepted, reserve funds, run a short move transition, save, and enter Tier 2 atomically.

The Garage Phase is complete only when the player can repeatedly make decisions, survive, learn, and legitimately reach this offer. Do not weaken its gate merely to expose employees.

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6. Tier 2: First Office

6.1 Identity

This is the player's first transformation from solo creator to team leader. The new challenge is not "more bubbles." It is choosing people, paying them, onboarding them, assigning work, balancing design and technology, and surviving the higher burn rate.

6.2 Capacity

- 4 HQ seats total.
- Seat 1 is permanently the founder.
- Seats 2-4 may hold three hired production employees.
- One normal game project may be active.
- One very small auxiliary activity may run only if assigned capacity remains; examples are a short report, minor patch, or basic research task.
- No R&D or Hardware remote slots.

6.3 Unlocks on move completion

- Candidate search and hiring.
- Employee roster.
- Payroll and employment ledger entries.
- Welcome Training.
- Basic books/practice training.
- Assignment view for the nine production disciplines.
- Team cohesion and onboarding.
- Employee energy, fatigue, morale, vacation, and burnout risk.
- Medium Games research.
- Basic publishing deals for Medium games.
- Better contracts.
- Basic marketing, when era/capability requirements are met.
- General staff-generated RP.

6.4 Hiring flow

1. Select a search method:
   - Algorithms: biases Technology and Research.
   - Showreel: biases Design and creative disciplines.
   - Game Demo: biases balanced candidates.
2. Select search budget and duration.
3. Generate a deterministic candidate slate from campaign seed, market, budget, and date.
4. Inspect Design, Technology, Speed, Research, salary request, personality modifiers, preferred disciplines, and hidden potential range.
5. Make an offer or close the slate.
6. An accepted candidate enters `onboarding`, not full effectiveness.
7. Welcome Training reduces ramp time and early cohesion penalty.
8. Every hire writes salary, signing, and hiring-cost ledger entries.

Candidates do not reroll when the same slate is reopened. Paying for a new search creates a new deterministic search event.

6.5 Team onboarding and cohesion

New staff create a real but bounded coordination cost:

```
newHireEfficiency = 0.55 + 0.25 * welcomeTrainingCompletion
teamCohesion = weightedAverage(pairFamiliarity, tenure, morale, leadershipSupport)
```

- New hires begin between 55% and 80% contribution.
- Normal work, completed projects, welcome training, and compatible assignments increase cohesion.
- Firing, rapid hiring, crunch, missed payroll, and repeated project cancellation reduce cohesion.
- Cohesion affects coordination loss and bug risk, not raw concept fit.
- A good topic/genre combination cannot erase a dysfunctional team.

6.6 Basic training

Training consumes the selected person's time. It cannot occur while that person contributes full work to a project.

Initial Tier-2 categories:

- Welcome Training.

- Design foundations.
- Technology foundations.
- Speed/practice.
- Research methods.
- Management Fundamentals for the founder.
- Introductory discipline workshops.

Training has diminishing returns when spammed without practical work. The exact cooldown belongs in configuration and is visible in Analyst Mode.

6.7 General staff RP

Only active hired production employees count. Founder activity RP remains separate.

```
weeklyStaffRp = eligibleProductionEmployees * 1
```

An employee who spent most of the week on vacation, unpaid leave, or onboarding below the configured participation threshold does not produce a full point. Fractional RP is stored; the normal HUD may display whole RP.

6.8 Tier-2 game scale

Size	Availability	Recommended active people	Commercial path
Small	Existing	1-2	Self-publish
Medium	Research-gated	2-3	Publisher recommended until audience/fan reach supports self-publishing
Large	Hidden	-	Not available
AAA	Hidden	-	Not available

The player may assign all four people to a Medium game, but coordination and underutilization penalties make "more people" an actual planning choice rather than a guaranteed improvement.

6.9 Tier-2 completion proof

The office-upgrade offer appears only after the team demonstrates that it can function:

- All three employee seats are filled.
- The company has released at least four games since moving.
- Team cohesion is at least 70%.
- Management Fundamentals is complete.
- Cash and runway gates pass.
- The office has been occupied for at least 52 weeks.

The upgrade is an in-place renovation. No game or employee is deleted, moved to storage, or silently reassigned.

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7. Tier 3: Upgraded Office and Training Center

7.1 Identity

The background remains recognizably the same first office, but the company now has better equipment, a dedicated training destination, stronger management tools, and one additional production seat. This tier turns a group of hires into a deliberately shaped team.

7.2 Capacity

- 5 HQ seats total.

- Founder + four production employees.
- No lab remote slots.
- One main game project.
- One auxiliary project when separately staffed: engine integration, research, report, port, patch, contract, or training cohort.
- The auxiliary project is constrained by actual work capacity; opening another screen does not create free concurrency.

7.3 Unlocks

- Fifth HQ seat.
- Training Center.
- Advanced courses.
- Mentorship.
- Career plans.
- Retraining and controlled specialization changes.
- Team-lead assignments.
- Discipline specialization path for the nine production areas.
- Multi-Genre research when era requirements are met.
- Sequels and franchise tracking.
- Medium self-publishing when reach requirements are met.
- Large-game research and publisher-funded Large prototypes.
- Stronger marketing and convention options.
- Two-person engine integration without stopping all company activity.

7.4 Advanced training structure

Every employee retains four foundational stats:

- Design.
- Technology.
- Speed.
- Research.

Do not add a fifth all-purpose "Leadership" bubble. Leadership is a separate competency record earned through lead assignments, mentorship, project outcomes, and management training:

```
type LeadershipState = {
  managementXp: number
  certificationLevel: 0 | 1 | 2 | 3
  successfulMilestonesLed: number
  failedMilestonesLed: number
  directReportsSupported: number
}
```

Training Center paths:

1. Foundation: improve one of the four base stats.
2. Discipline: move toward Engine, Gameplay, Story & Quests, Dialogue, Level Design, AI, World Design, Graphics, or Sound specialization.
3. Leadership: project planning, feedback, estimation, mentoring, and risk management.
4. Cross-training: reduce severe mismatch penalties in a secondary discipline.
5. Retraining: replace a specialization through a costly, time-consuming transition without deleting the employee.

7.5 Specialization behavior

- A specialization increases fit, work quality, estimation accuracy, and defect detection in its discipline.
- It does not multiply every contribution from that employee.
- A specialist assigned far outside the specialty still contributes according to base skills and secondary mastery.
- Specializations belong to the employee and leave if the employee leaves.
- Knowledge produced by completed work remains in the studio knowledge base.
- Retraining temporarily removes the active specialization bonus and creates a new mastery ramp.

7.6 Mentorship

Mentorship pairs one senior with one junior for a defined period:

- The senior loses some immediate production capacity.
- The junior gains mastery and cohesion faster.
- Both gain small Research and leadership progress.
- Mentorship cannot be toggled repeatedly for free.
- A mentor may supervise only the configured number of mentees.

7.7 General staff RP

```
weeklyStaffRp = eligibleProductionEmployees * 2
```

At full staffing, the four hired production employees can generate up to 8 general RP per fully active week. The founder is not included in that count. Founder activity RP still applies when the founder develops, researches, reports, or performs another qualifying action.

7.8 Large-game bridge

Tier 3 introduces Large games carefully:

1. Research Large Production Planning.
2. Complete a publisher-funded Large prototype or vertical slice.
3. Demonstrate sufficient work capacity and an engine version that supports the selected features.
4. Develop a Large game under a publisher or with severe self-publishing risk until the studio's reach is high enough.

This produces the proof needed for the Technology Park without granting AAA early.

7.9 Technology Park proof

The move offer requires:

- Five occupied HQ seats.
- At least 12 total releases.
- At least three profitable Medium-or-larger games.
- At least one trained discipline specialist.
- Positive trailing 52-week operating profit.
- At least 104 weeks in Tier 3.
- $16M liquid cash and safe post-move runway.

The player may delay the move to train more, build cash, or finish a project. The Technology Park cannot be forced during active release settlement.

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8. Cross-tier player progression summary

System	Garage	First Office	Upgraded Office	Technology Park	Expanded Campus
HQ seats	1	4	5	6	8
Founder	Solo creator	Creator/manager	Creative lead/manager	CEO/creative lead	CEO/creative lead
Normal hires	None	3	4	5	5 production + 2 directors
Remote lab staff	None	None	None	2 per built lab	4 per lab
Basic training	Founder learning	Yes	Yes	Yes	Yes
Advanced training	No	No	Yes	Yes	Yes
Discipline specialists	No	Intro path	Yes	Required for scale	Mature career system
Small games	Yes	Yes	Yes	Yes	Yes
Medium games	No	Research	Yes	Yes	Yes
Large games	No	No	Publisher bridge	Yes	Yes
AAA	No	No	No	R&D path	Mature production class
MMO	No	No	No	Basic path	Mature persistent-service business
MMORPG/subscriptions	No	No	No	Research groundwork	Full system
Custom engines	Basic	Team engines	Versioned pipeline	Advanced architecture	Engine licensing/ecosystem
R&D Lab	Hidden	Hidden	Hidden	Design Specialist unlock	Dedicated director/division
Hardware Lab	Hidden	Hidden	Hidden	Tech Specialist + R&D unlock	Dedicated director/division
Normal game concurrency	1	1	1 + auxiliary	1 main + labs/support	1 main + 1 secondary + labs/live ops, subject to capacity
Post-launch support	Basic patch	Assigned support	Auxiliary support	Dedicated capacity tradeoff	Managed live-operations portfolio

9. Tier 4: Technology Park

9.1 Identity

The Technology Park is the first true major-studio environment. It should feel like the player has arrived without implying the progression is over. The production group can now tackle Large work reliably, prove AAA capability, and establish R&D and Hardware divisions.

This tier deliberately creates a leadership bottleneck. The company can build both labs, but until the campus expands, qualified HQ specialists must act as part-time lab leads. A lab therefore competes with production for leadership attention. The Tier-5 expansion solves that bottleneck through two dedicated director seats.

9.2 Move transaction

On acceptance:

1. Validate cash, runway, staffing, active-project safety, and offer identity.
2. Reserve the $8M move cost.
3. Finish or safely pause incompatible office-bound activities.
4. Preserve every employee, assignment, project, engine, game, research item, contract, and ledger entry.

5. Create the Technology Park office state with six HQ seats.
6. Keep the five existing people assigned; leave the sixth seat vacant.
7. Unlock hiring for the sixth seat.
8. Reveal the "Advanced Divisions" destination only after the move completes.
9. Start the independent `technologyParkTenureWeeks` counter.
10. Save immediately after the atomic move.

9.3 Capacity

- 6 HQ seats total.
- Founder + five normal production employees at full staffing.
- One main production project.
- One auxiliary production/support project when staffed.
- One R&D Lab project slot after that lab is built.
- One Hardware Lab project slot after that lab is built.
- Two remote staff slots per built lab.
- One active live-service operation slot after the required research.

Concurrency is not permission to ignore work capacity. A six-person AAA project can leave no people for a port, report, engine integration, or acting lab leadership.

9.4 General staff RP

```
weeklyStaffRp = eligibleProductionEmployees * 4
```

At full staffing, the five hired production employees can generate up to 20 general RP per fully active week. The founder and remote lab staff are excluded. An HQ employee spending most of the week as an acting lab lead receives lab progress and only the prorated general RP tied to actual production participation.

9.5 Technology Park unlocks

- Sixth HQ seat.
- Reliable Large self-publishing when reach permits.
- Large multi-platform development.
- Advanced engine architecture and performance budgets.
- Design Specialist and Technology Specialist career tracks.
- R&D Lab construction path.
- Hardware Lab construction path.
- AAA Production research path.
- Network Infrastructure research path.
- Basic persistent online-game path.
- Engine-licensing groundwork.
- Advanced conventions and major marketing campaigns.
- Larger publishing negotiations.
- Dedicated post-release assignments.

None of the labs, AAA, MMO, or hardware systems becomes usable merely because the background changed.

10. R&D Lab: unlock, staffing, and function

10.1 Unlock sequence

The sequence mirrors the original's dependency logic but uses Studio Empire's versioned research model:

1. Own the Technology Park.
2. Train any eligible founder or HQ employee into a Design Specialist.
3. Discover the R&D Lab proposal.
4. Review construction cost, weekly base overhead, remote slots, and acting-lead requirement.
5. Build the lab through a timed office project.
6. Assign the Design Specialist as acting R&D Lead.
7. Hire up to two remote R&D staff.
8. Fund a research portfolio or allow the lab to generate foundational RP/knowledge.

10.2 Design Specialist eligibility

Initial tuning values:

- Employee level 7 or equivalent career maturity.
- Design >=700.
- Research >=400.
- At least two successful projects at Medium size or larger.
- Leadership Certification 1 for an acting-lead assignment.
- Specialist training: 100 RP, $5M, and configured calendar time.

The founder may qualify, but assigning the founder as acting lead prevents the founder from simultaneously providing full production contribution.

10.3 R&D staff model

```
type LabStaffAssignment = {
  employeeId: string
  labId: "rd" | "hardware"
  allocation: number // 0.0-1.0
  role: "acting_lead" | "director" | "researcher" | "engineer" | "analyst"
}
```

- The acting lead occupies an existing HQ seat and divides capacity between HQ and lab work.
- Remote R&D staff occupy lab slots, not HQ seats.
- Remote staff remain individually hireable, trainable, promotable, fireable, and saveable.
- No remote lab employee appears as a full-size main-scene sprite.
- Remote lab staff create lab work and lab knowledge, not normal game-production work unless explicitly transferred and assigned through a supported command.

10.4 R&D work types

The lab supports four distinct activities:

1. Foundational research: discovers a capability or raises scientific maturity.
2. Prototype: tests whether a discovered technology is viable.
3. Engine transfer: turns a successful prototype into an integration project for a new engine version.
4. Major project: creates studio-level capabilities such as advanced online infrastructure, AAA production processes, engine licensing, or a distribution network.

Research never edits an active engine or game in place.

10.5 R&D portfolio

The player selects:

- One primary project.
- Optional background research theme.
- Weekly budget.
- Risk posture: conservative, balanced, or experimental.
- Acting-lead allocation.
- Remote-staff assignments.

Higher funding increases available equipment and experimentation, but returns diminish. It cannot buy instant completion.

```
weeklyRdWork = staffWork * leadModifier * equipmentModifier * maturityModifier
equipmentModifier = 1 + diminishingReturnCurve(weeklyBudget / recommendedBudget)
```

10.6 When no major project is active

The lab does not become useless. It may:

- Produce a bounded amount of lab RP.
- Improve selected technology maturity slowly.
- Reduce unknowns on a discovered item.
- Mentor a qualifying employee.
- Analyze technical debt from released engines.

Idle generation must be capped and cost money. It cannot become an infinite RP printer with no opportunity cost.

10.7 Initial R&D major-project tree

```
Advanced Production Methods
├── AAA Production
│   ├── Milestone Planning
│   ├── Advanced QA
│   └── Global Launch Operations
├── Internet Infrastructure
│   ├── Persistent Servers
│   │   ├── MMO Architecture
│   │   └── Live Operations
│   └── Digital Distribution
│       ├── Engine Licensing
│       └── Studio Storefront
└── Hardware Architecture
    └── Hardware Lab proposal
```

Later Tier-5 branches extend this tree. Do not show them while their systems are unfinished.

---

11. Hardware Lab: unlock, staffing, and function

11.1 Unlock sequence

1. Own the Technology Park.
2. Build the R&D Lab.
3. Complete R&D's Hardware Architecture major project.
4. Train any eligible founder or HQ employee into a Technology Specialist.
5. Discover the Hardware Lab proposal.
6. Build the lab through a timed office project.
7. Assign the Technology Specialist as acting Hardware Lead.
8. Hire up to two remote Hardware employees.

11.2 Technology Specialist eligibility

Initial tuning values:

- Employee level 7 or equivalent maturity.
- Technology >=700.
- Research >=400.
- At least two successful Medium-or-larger projects.
- Leadership Certification 1 for an acting-lead assignment.
- Specialist training: 100 RP, $5M, and configured calendar time.

11.3 Hardware Lab work types

1. Component and architecture research.
2. Development kits.
3. Platform prototypes.
4. Manufacturing and reliability planning.
5. Game-specific hardware optimization.
6. Custom hardware/platform projects.
7. Revisions, cost reductions, and successor planning.
8. Platform support and retirement.

The lab must not be a second R&D point bar with a different icon. It manages physical constraints, unit economics, reliability, compatibility, manufacturing, and installed-base strategy.

11.4 Acting-lead tradeoff

At Tier 4, an acting Hardware Lead and acting R&D Lead may be different HQ employees. If both labs run major projects while a six-person AAA game is active, the company is overcommitted. The simulation must show the shortage before acceptance and then enforce its consequences if the player proceeds.

Possible player responses:

- Reduce a lab budget and pause its project at a safe milestone.
- Delay the AAA game.
- Use fewer people on the game and accept schedule/risk impact.
- Hire remote lab staff to reduce, but not eliminate, acting-lead demand.
- Finish one project before starting the other.
- Wait for the Expanded Campus and promote dedicated directors.

---

12. Tier 4 AAA Production

12.1 AAA is a production class, not a size button

AAA becomes discoverable only after the studio demonstrates Large-game excellence. Initial proof:

- R&D Lab exists.
- At least one Large game receives a displayed review average of 9.5 or better, or two Large games receive 9.0 or better.
- At least one Large game is profitable after full development and marketing costs.
- The selected engine family supports the configured advanced graphics/performance generation.
- At least three employees have relevant discipline specializations.
- The company is not in insolvency.

The proof unlocks the AAA Production R&D project. Completing that project unlocks AAA preproduction; it does not create a game automatically.

12.2 AAA project state machine

```
concept
-> greenlight_planning
-> vertical_slice
-> preproduction_review
-> stage_1
-> milestone_1_review
-> stage_2
-> milestone_2_review
-> stage_3
-> content_lock
-> alpha
-> beta
-> certification
-> release_candidate
-> release | cancel
-> launch_support
-> normal_support | live_service | sunset
```

Every transition is domain-owned and saved. A UI modal cannot skip a milestone.

12.3 Greenlight package

Before AAA work starts, the player approves:

- Concept, target audience, genre, topic, platforms, and engine version.
- Total target scope and nine discipline intents.
- Core team assignments.
- Milestone schedule.
- Base budget and contingency reserve.
- Marketing ceiling, separate from quality work.
- QA plan.
- Outsourcing/contractor allowance if that system is enabled.
- Cancellation checkpoints.
- Post-launch support plan.

The screen must show projected work, not promise a review score.

12.4 Vertical slice

The vertical slice tests the riskiest connected portion of the game. It can reveal:

- Engine-feature mismatch.
- Performance risk.
- Unproven gameplay loop.
- Content-production bottleneck.
- Platform-build risk.
- Staff overload.
- Schedule underestimation.

The player may revise scope, add time, change supported features before lock, or cancel. A strong vertical slice reduces uncertainty; it does not add a magical quality bonus.

12.5 Milestone decisions

At each milestone the player sees:

- Completed versus required work.

- Known and estimated unknown bugs.
- Technical debt.
- Team energy and morale.
- Schedule variance.
- Budget variance.
- Platform risk.
- Quality confidence bands, not final review scores.

Allowed decisions include rescope, delay, reassign, increase QA, approve overtime, pause, or cancel. Overtime creates short-term work at the cost of fatigue, morale, defect probability, and retention. It is never a free speed multiplier.

12.6 AAA lab support

During active AAA production:

- R&D may run a marketing/technology support project that improves tooling, risk reduction, or launch awareness according to its type.
- Hardware may run a platform-optimization or showcase project that improves performance/certification confidence or market reach on the relevant hardware.
- Neither lab directly changes review quality after production truth is determined.
- A lab support project consumes its normal people, money, and time.

12.7 AAA completion and post-launch

AAA release creates larger support obligations:

- Day-one patch decision.
- Platform certification issues.
- Regional launch support.
- Critical bug response.
- Content roadmap promises.
- Community expectations.

A successful AAA game may become a normal long-tail title, a franchise base, or a live-service candidate. "AAA" and "live service" remain independent concepts.

---

13. Basic online and MMO path at Tier 4

13.1 Dependency sequence

```
Internet Infrastructure
-> Persistent Servers
-> Online Accounts
-> Live Operations Fundamentals
-> MMO Architecture
```

An MMO is a service model, not a genre. An RPG MMO becomes an MMORPG only when the project also uses RPG genre expertise and unlocks the persistent-world RPG feature set.

13.2 Tier-4 limitations

Tier 4 may operate one persistent online title at a time. It supports:

- Premium online games.
- Persistent multiplayer worlds.
- A basic MMO launch.

- Server budget and capacity.
- Patches.
- One expansion/content update at a time.
- Basic community health.
- Shutdown/take-off-market decision.

Tier 4 does not yet support the full subscription portfolio, multiple live-service operations, advanced monetization, cross-game accounts, or a company-wide online platform. Those require Tier 5 and the dedicated R&D Director.

13.3 Required live-service assignment

An active persistent title consumes Operations Capacity every week. If nobody is assigned:

- Bugs and incidents resolve slowly.
- Content freshness decays.
- Community trust falls.
- Churn rises.
- Server cost continues.

The player cannot release an MMO, reassign everyone to a new AAA title, and collect risk-free passive money.

---

14. Tier 5: Expanded Technology Campus

14.1 Unlock rule

The expansion opportunity is discovered after exactly 12 full campaign years of Technology Park tenure:

```
expandedCampusTenureRequirementWeeks = 12 * 52
```

Pause, speed, office screen visits, lab construction, and staff changes do not alter this counter. A save/load cycle cannot reset or advance it. The counter starts only when the Technology Park move completes.

At the threshold:

- Queue one campus-expansion proposal.
- Show that the additional two HQ seats are specialized leadership seats.
- Show expanded remote-lab capacity.
- Show construction cost, overhead, downtime, and runway.
- Do not require a Level-7 graphics technology or specific game score.
- Let the player decline and reconsider indefinitely.

14.2 Capacity after completion

- 8 HQ seats total.
- Six-person production group remains: founder + five production employees.
- One R&D Director seat.
- One Hardware Director seat.
- Up to four remote R&D staff.
- Up to four remote Hardware staff.
- One main game project.
- One secondary production project when capacity permits.
- One R&D major project.
- One Hardware major project.
- Up to three persistent-service operation slots, constrained by operations staffing and infrastructure.

The extra two seats do not increase normal production-team size from six to eight. They remove the acting-lead tax and create specialized divisional management.

14.3 Expansion construction behavior

Expansion is a project with phases:

```
proposal -> funding_reserved -> planning -> construction -> systems_setup -> completed
```

- Existing studio work may continue during planning.
- Construction creates configured noise/disruption overhead but should not arbitrarily freeze the company for 12 years.
- Final systems setup may require a short reduced-capacity window.
- If the company enters insolvency before completion, construction may pause; reserved and spent funds remain auditable.
- Cancellation refunds only the unspent portion and retains sunk cost.
- Completion adds empty director seats. It does not conjure qualified directors.

14.4 Director vacancy behavior

The company may finish the expansion with one or both director seats vacant.

- Existing lab projects may continue to the next safe milestone under an acting lead.
- No new Tier-5 major project may start without the correct director.
- Remote staff cap remains at Tier-4 effective capacity until a director is appointed.
- The relevant lab runs with reduced throughput and higher coordination loss.
- A director vacancy is an operational state, not a game-over condition.

14.5 Director appointment paths

The player may:

1. Promote an existing qualified HQ employee.
2. Promote a qualified remote lab employee.
3. Run an executive external search.

Promotion preserves the person, history, mastery, morale, relationships, and salary negotiation. Moving a production employee into a director seat opens a normal production seat that may be rehired.

External directors have stronger immediate leadership but weaker company familiarity and require onboarding.

---

15. R&D Director role

15.1 Eligibility

Initial tuning requirements:

- Design Specialist certification.
- Research >=700.
- Design >=750.
- Leadership Certification 2.
- At least three successful major milestones or completed research projects.
- No active burnout state.
- Executive training completion.

15.2 Direct job functions

The R&D Director is not a decorative manager. The role owns:

- Quarterly research portfolio proposal.
- Primary and background research themes.
- Lab budget recommendation.
- Project-risk posture.
- Remote R&D hiring and assignment recommendations.
- Prototype reviews.
- Technology-transfer readiness.
- Engine-integration handoff quality.
- Research debt and abandoned-prototype review.
- Mentorship of Design/Research specialists.
- Live-service architecture and software-platform roadmap.

The player retains final authority. The director converts complex raw lab data into recommendations and reduces coordination loss; the director does not play the game automatically.

15.3 Director contribution

The director contributes leadership and technical review to R&D. The director may assist a game at a milestone review, but cannot simultaneously count as a full production worker.

```
rdDirectorModifier =
  leadershipFit
  * labFamiliarity
  * moraleModifier
  * allocationModifier
```

The modifier affects estimation accuracy, rework, risk discovery, and lab efficiency. It does not multiply final game reviews.

15.4 Failure, firing, and replacement

- Firing or losing the director pauses new R&D starts.
- Active work reaches a safe checkpoint or enters a documented paused state.
- Studio knowledge and completed research remain.
- Personal mastery, relationships, and unrecorded insight leave with the employee.
- A replacement incurs handoff/ramp cost based on documentation maturity.
- The player may appoint a temporary acting lead from HQ with the old production-capacity tradeoff.

---

16. Hardware Director role

16.1 Eligibility

Initial tuning requirements:

- Technology Specialist certification.
- Research >=700.
- Technology >=750.
- Leadership Certification 2.
- At least three successful hardware, engine, performance, or platform milestones.
- No active burnout state.
- Executive training completion.

16.2 Direct job functions

The Hardware Director owns:

- Hardware/platform roadmap proposals.
- Architecture and component tradeoffs.
- Reliability target.
- Manufacturing partner recommendations.
- Unit-cost and launch-price scenarios.
- Development-kit readiness.
- Backward-compatibility strategy.
- Platform certification policy.
- Remote Hardware hiring and assignments.
- Revision and successor timing.
- Supply-risk and recall response.
- First-party launch-lineup coordination with production.

16.3 Director contribution

The Hardware Director improves architecture evaluation, vendor-risk visibility, cost estimation, reliability discovery, and hardware-team coordination. The role does not directly create market demand. A technically excellent console still needs a good price, launch lineup, developer support, and audience fit.

16.4 Failure, firing, and replacement

The same safe-pause and knowledge-retention rules used by R&D apply. Active manufacturing cannot vanish; support obligations and vendor contracts remain even if the director leaves.

---

17. Vertical-screen office and remote-work presentation

17.1 Non-negotiable visual rule

Only one full-size character is rendered in the main office scene at a time. This applies in every tier. Additional people are real simulation entities represented through roster chips, presence indicators, activity messages, and dedicated management destinations.

The simulation must never reduce staff contribution merely because a sprite is not on screen.

17.2 Focus-character rules

```
type FocusCharacterReason =
  | "player_selected"
  | "current_primary_activity"
  | "milestone_lead"
  | "critical_event"
  | "new_hire_welcome"
  | "director_update"
```

Priority:

1. Blocking critical event participant.
2. Player-selected employee.
3. Current project's milestone lead.
4. Founder.

Changing the focus character is visual only. It cannot reassign work or alter output unless the player explicitly dispatches an assignment command from the character card.

17.3 Main-scene elements

Keep the room as a place, not a dashboard. It may contain:

- One focus character and one active animation.
- Studio name, campaign date, industry era, cash, fans, and current primary action.
- A compact remote-team strip.
- Contextual destinations such as Games, Research, Staff, Training, Contracts, Finances, R&D, and Hardware when unlocked.
- Ambient state changes tied to real events.
- One high-priority notification at a time.

Do not place full staff spreadsheets, nine development sliders, lab budgets, and sales tables permanently on the room.

17.4 Remote-team strip

Each off-screen employee has a compact chip with:

- Portrait.
- Name.
- Current assignment icon.
- Energy/morale warning only when actionable.
- Progress ring for the current work item.
- Remote/on-site status for flavor and scheduling.
- Critical status such as vacation, burnout, onboarding, or idle.

Tapping a chip focuses that employee and opens their concise card. A second action opens the full roster or assignment screen.

17.5 Remote work is presentation plus a bounded mechanic

Remote status may affect:

- Commute/availability flavor.
- Collaboration overhead for poorly documented teams.
- Morale preferences.
- Office disruption resilience.
- Hiring pool geography.

It must not become a universal penalty. Documentation, cohesion, leadership, tooling, and employee preference determine the effect. The game should make a well-run remote studio viable.

17.6 Tier backgrounds

Tier	Main visual evolution
Garage	One desk, car/storage, early equipment, founder alone
First Office	Small professional room, remote-team panel, empty/fillable seat cues without rendering all staff
Upgraded Office	Same recognizable room with improved hardware and Training destination
Technology Park	Larger campus office, visible but inaccessible lab wings until built
Expanded Campus	Same campus expanded; R&D and Hardware director destinations become distinct; future-tech cues remain restrained

The change from Tier 2 to Tier 3 must look like an upgrade to the same place, not a teleport to a different building.

---

18. Complete employee lifecycle

18.1 Employee states

```
type EmploymentState =
  | "candidate"
  | "offer_pending"
  | "onboarding"
  | "active"
  | "training"
  | "vacation"
  | "leave"
  | "burned_out"
  | "notice_period"
  | "terminated"
  | "departed"
```

Every transition writes a dated event. Departed employees remain in history; their IDs are never reused.

18.2 Stats and persistent growth

Hired employees have:

- Design.
- Technology.
- Speed.
- Research.
- Level/career maturity.
- Nine discipline masteries.
- Leadership state.
- Energy.
- Fatigue.
- Morale.
- Burnout risk.
- Cohesion relationships.
- Salary and employment terms.
- Training history and cooldowns.
- Project, genre, topic, engine, and platform experience.
- One active primary assignment and bounded secondary support allocation.

18.3 Hire

- Candidate creation is deterministic.
- Search method, budget, era, reputation, office tier, and company culture influence the pool.
- Better candidates demand more salary or stronger terms.
- Famous/rare candidates may exist but are never required for progression.
- Hiring into a full seat is blocked unless a pending departure creates a dated vacancy.
- Remote lab candidates cannot be hired before the lab and slot exist.

18.4 Fire or lay off

The player may terminate staff at any time outside an atomic milestone/release transaction, but consequences apply:

- Severance and contract cost.

- Morale impact on related staff.
- Cohesion disruption.
- Loss of that employee's personal mastery.
- Open assignments become understaffed.
- Director loss invokes lab continuity rules.
- Studio knowledge already captured in reports, engines, prototypes, and completed projects remains.

No moralizing popup is required. The simulation should make the decision legible and consequential.

18.5 Retrain

The player can rebuild a team rather than permanently discarding people:

1. Select a new target specialization or career path.
2. Show retained transferable mastery and lost active specialization bonus.
3. Show money, RP, time, and temporary-capacity cost.
4. Begin retraining.
5. Employee remains paid and may not contribute full production work.
6. On completion, activate the new specialization at an initial mastery level.

This remains available in Endless Mode, so the player can fire, hire, promote, or retrain an entire roster after the scored campaign.

18.6 Energy, vacation, morale, and burnout

- Energy affects hired-employee contribution.
- Normal recovery occurs off assignment and during vacation.
- Repeated overtime, crunch, failed milestones, and ignored vacation increase burnout risk.
- Burnout reduces contribution and may trigger leave or departure.
- Salary fairness, project success, training opportunity, preferred work, remote preference, leadership quality, and job security affect morale.
- The founder still does not receive the hired-employee energy system.

18.7 Knowledge versus people

This separation is mandatory:

- Personal mastery leaves with the person.
- Documented studio knowledge remains.
- Engine versions and source assets remain.
- Unfinished undocumented insight may be partially lost.
- Research completion remains; project velocity may fall after expert departure.

This prevents both extremes: firing everyone should hurt, but it must not erase decades of company history.

---

19. Research Points and learning across tiers

19.1 RP and skill XP remain separate

- RP is a studio resource spent on research and training.
- Skill XP/mastery belongs to people and improves what they can do.
- Lab maturity belongs to the studio/division.
- Engine maturity belongs to an engine family/version lineage.

19.2 Founder activity RP

Retain the existing initial rates unless balance tests replace them:

Founder activity	RP per active day
Developing/polish	0.15
Researching	0.25
Writing Game Report	0.20
Contract work	0.10

Fractional RP persists. Reopening a screen cannot award the same day twice.

19.3 Passive market learning

For each non-dormant released title:

```
weeklyMarketRp = 0.4 * qualityFactor * sizeFactor
qualityFactor = clamp(avgReview / 7, 0.5, 1.4)
sizeFactor = small 1.0 | medium 1.25 | large 1.5 | aaa 1.8
```

This represents telemetry, support experience, and market learning. A flop teaches a little; a hit teaches more. Dormant or fully sunset games stop normal weekly learning.

19.4 Production-employee weekly RP

Tier	Eligible hired production employees at capacity	RP per eligible employee/week	Maximum base weekly RP
Garage	0	-	0
First Office	3	1	3
Upgraded Office	4	2	8
Technology Park	5	4	20
Expanded Campus	5	4	20

Directors and remote lab staff generate lab work and bounded lab RP according to lab rules. They never inflate the normal production-employee formula.

19.5 Anti-farming rules

- No RP for opening/closing UI.
- No RP for paused time.
- No duplicate daily or weekly settlement.
- Repeating trivial research/engine projects has diminishing learning.
- Abandoning a project may yield partial documented learning but never the full completion reward.
- A released game cannot produce passive RP after it is sunset merely because its history card remains.
- General RP generation is capped by real participation, not nominal headcount.

---

20. Game-size progression and project demand

Game size determines scope class, coordination, budget, and commercial expectations. It does not guarantee quality or sales.

20.1 Initial size targets

Size	First normal tier	Recommended core staff	Typical duration target	Minimum management structure	Normal post-launch burden
Small	Garage	1-2	28-42 simulation weeks for an early founder game; faster with mature tools	Single owner	Low
Medium	First Office	2-3	35-70 weeks	Project lead	Moderate
Large	Upgraded Office bridge / Technology Park normal	4-6	70-130 weeks	Lead + milestone planning	High
AAA	Technology Park via R&D	Six-person production group plus optional lab/contract support	120-240 weeks	Greenlight, leads, milestones, QA, certification	Very high

These ranges are balance targets around Standard Work Units. The calendar outcome comes from required work divided by real capacity, not a hard timer.

20.2 Scope pressure

```
requiredSWU =
  baseSizeSWU
  * disciplineIntentDemand
  * selectedFeatureDemand
  * platformBuildDemand
  * noveltyDemand
  * qualityTargetDemand

capacitySWU = sum(assignedPersonWeeklyContribution)

schedulePressure = requiredSWU / max(1, capacitySWU * plannedWeeks)
```

High schedule pressure increases estimate uncertainty, defects, rework, fatigue, and missed milestones. It never directly subtracts a mysterious review point.

20.3 Team-size misuse

- Too few people creates schedule and workload risk.
- Too many people on a small scope creates coordination loss and underutilization.
- Untrained teams suffer more coordination loss than cohesive teams.
- Better leadership, tooling, documentation, and office tier raise manageable complexity but never remove it.

20.4 Self-publishing reach

Publisher access and self-publishing remain choices. Initial reach guidelines:

Size	Recommended audience reach for self-publishing without severe discoverability risk
Small	No minimum
Medium	100,000 fans/reachable audience equivalent
Large	250,000
AAA	1,000,000

Falling below these thresholds changes market reach and financial risk, not execution quality or review truth. A great unknown game can become a slow-burn hit; a huge campaign can make a mediocre game spike and collapse.

21. Persistent online games, MMO, and MMORPG

21.1 Product taxonomy

Keep these separate:

Concept	Meaning
Online feature	A feature such as leaderboards, co-op, or matchmaking
Persistent online game	Retains accounts, state, or service dependencies between sessions
Live-service game	Receives ongoing operations/content and tracks an active audience
MMO	Supports a large shared service population; not a genre
MMORPG	MMO service model + RPG genre expertise + persistent-world RPG systems
Subscription	A monetization model; may apply to an MMO, a game catalog, or a platform service

The creation screen must not place "MMO" beside Action/RPG/Strategy as though they are equivalent genre choices.

21.2 Unlock graph

```
Internet Infrastructure
├── Online Accounts
├── Matchmaking
└── Persistent Servers
    ├── Live Operations Fundamentals
    │   ├── Content Updates
    │   ├── Community Management
    │   └── Service Analytics
    └── MMO Architecture
        ├── Sharded Worlds
        ├── Persistent Economy
        ├── Guild/Social Systems
        └── MMORPG Framework (also requires RPG mastery)

Expanded Campus + R&D Director
├── Subscription Billing
│   ├── Premium Subscription
│   ├── Free Trial
│   └── Catalog Subscription
├── Cross-Platform Identity
├── Global Service Fabric
├── User-Generated Content
└── Cloud-Native Worlds
```

21.3 Service project state machine

```
type ServiceLifecycle =
  | "not_applicable"
  | "prelaunch_capacity_planning"
  | "closed_alpha"
  | "closed_beta"
  | "open_beta"
  | "launching"
  | "live_growth"
  | "live_stable"
  | "live_decline"
  | "maintenance_mode"
  | "sunset_announced"
  | "sunset"
```

The normal game-project lifecycle still creates the product. The service lifecycle begins when persistent infrastructure is approved. Both states must be saved and linked.

21.4 Core service metrics

```
type LiveServiceState = {
  gameId: string
  activeUsers: number
  peakConcurrentUsers: number
  subscribers: number
  trialUsers: number
  conversionRate: number
  weeklyRetention: number
  weeklyChurn: number
  contentFreshness: number
  communityTrust: number
  serverCapacity: number
  serverReliability: number
  technicalDebt: number
  operationsCapacityRequired: number
  operationsCapacityAssigned: number
  currentSeasonOrExpansionId?: string
  monetizationModel: MonetizationModel
}
```

21.5 Weekly service settlement

The order is fixed:

1. Settle incidents and reliability.
2. Apply content-freshness decay or update gains.
3. Calculate new user awareness/reach.
4. Calculate capacity and onboarding success.
5. Calculate retention and churn.
6. Calculate active users and subscribers.
7. Calculate revenue by monetization model.
8. Calculate infrastructure, staffing, licensing, support, and content costs.
9. Update community trust, franchise reputation, and operations knowledge.
10. Emit only actionable events; send raw details to history/analytics.

21.6 Initial formulas

```
capacityHealth = clamp(serverCapacity / max(1, demandAtPeak), 0, 1.25)

serviceQuality = weightedGeometricMean(
  launchExecutionQuality,
  serverReliability,
  contentFreshness,
  operationsCoverage,
  communityTrust
)

weeklyChurn = clamp(
  baseChurn
  + outagePenalty
  + staleContentPenalty
  + monetizationFriction
  + supportNeglectPenalty
  - socialRetentionBonus
  - updateBonus,
  minChurn,
  maxChurn
)

retainedUsers = priorActiveUsers * (1 - weeklyChurn)
newUsers = reachableMarket * awareness * conversion * capacityOnboarding
activeUsers = retainedUsers + newUsers
```

All factors are bounded and inspectable. No single hidden multiplier may turn a healthy service to zero without an explicit incident or shutdown.

21.7 Operations Capacity

Operations Capacity is produced by assigned staff, tools, lab infrastructure, and directors. It is consumed by:

- Active live titles.
- Incident response.
- Content seasons.
- Expansions.
- Community programs.
- Account/platform services.

If demand exceeds capacity, the player chooses what degrades or the system applies declared priorities. A company cannot operate unlimited MMOs because the screen allows unlimited history cards.

21.8 Content updates and expansions

Updates are real projects with scope, work, bugs, cost, and release states.

Update type	Typical purpose	Cost/risk	Effect
Maintenance patch	Reliability/bugs	Low	Restores trust and reliability
Feature update	New systems	Medium	Raises freshness; may add defects/debt
Season	Recurring content cadence	Medium	Temporary engagement and acquisition
Expansion	Major content	High	Large freshness/revenue opportunity; meaningful production demand
Engine migration	Technical longevity	Very high	Reduces obsolescence but risks outages/regressions

An expansion does not simply reset sales to launch week. Its effect depends on quality, relevance, audience size, service health, marketing reach, and fatigue.

21.9 Sunset

The player may announce shutdown. The flow includes:

- Proposed shutdown date.
- Remaining support obligations.
- Subscription billing stop/refund rules.
- Community-trust impact.
- Franchise/IP impact.
- Data preservation or migration decision.
- Final operating cost.

An emergency immediate shutdown is possible during insolvency but carries harsher consequences.

---

22. Subscription and monetization systems

22.1 Design principle

Monetization affects adoption, conversion, retention, revenue, reputation, and community trust. It never rewrites the actual production quality used by reviewers.

22.2 Models

```
type MonetizationModel =
  | "premium_purchase"
  | "premium_plus_subscription"
  | "subscription_only"
  | "free_trial_to_subscription"
  | "free_to_play_cosmetic"
  | "catalog_subscription"
```

Do not add manipulative dark-pattern mechanics as an optimal hidden strategy. Aggressive monetization may raise short-term revenue while increasing churn, review friction, regulation risk, and trust loss.

22.3 Premium subscription

Player choices:

- Monthly price.
- Included base content.
- Trial length.
- Expansion inclusion.
- Regional pricing abstraction.
- Billing/support staffing.

```
weeklySubscriptionRevenue = activeSubscribers * monthlyPrice * 12 / 52
```

Revenue then subtracts platform fees, payment fees, refunds, server costs, support, live-operations staffing, and content-production cost.

22.4 Catalog subscription

This is a company/platform service, not a checkbox on one game. It requires:

- Tier 5.
- R&D Director.
- Digital Distribution.
- Cross-Platform Identity.
- A minimum owned catalog or signed third-party content.
- Ongoing content, infrastructure, support, and partner payments.

Its core tension is catalog cost versus subscriber acquisition/retention. It may increase reach for included games while reducing unit sales. Those channels must be accounted separately.

22.5 Trust and fatigue

Repeated price increases, thin updates, broken launches, removed paid content, or abrupt shutdowns create account-level trust damage. Fair support, stable service, meaningful updates, and clear sunset handling rebuild trust slowly.

---

23. Custom hardware and platform ecosystem

23.1 Hardware is a platform business

A custom device is not complete when its research bar fills. It progresses through:

```
market_thesis
-> architecture
-> component_selection
-> prototype
-> developer_kit
-> manufacturing_plan
-> launch_lineup
```

```
-> production
-> launch
-> active_support
-> revision
-> successor_or_retirement
```

23.2 Market thesis

The player chooses:

- Form factor: home, handheld, hybrid, spatial/VR, cloud endpoint, or later future category.
- Target audience.
- Target price range.
- Performance tier.
- Developer friendliness.
- First-party strategy.
- Backward compatibility.
- Online/service strategy.
- Planned lifecycle.

The thesis creates constraints; it does not determine success.

23.3 Architecture and components

Configurable dimensions:

- Compute architecture.
- Graphics architecture.
- Memory/storage.
- Media/distribution.
- Input systems.
- Network capability.
- Backward compatibility.
- Accessibility capabilities.
- Power/thermal target.
- Reliability investment.
- Manufacturing complexity.

Every component has cost, performance, maturity, supply risk, integration work, and failure risk.

23.4 Prototype and reliability

Prototype testing discovers:

- Performance gaps.
- Thermal/power failures.
- Reliability defects.
- Cost overruns.
- Toolchain issues.
- Manufacturing yield risk.
- Developer-kit friction.

The player may revise the device, accept risk, delay, or cancel. Reliability spending has diminishing returns but cannot be reduced to zero without consequences.

23.5 Developer ecosystem

Platform adoption depends on:

- Device value and price.
- Launch lineup.
- First-party exclusives.
- Development-kit quality.
- Licensing terms.
- Installed-base momentum.
- Backward compatibility.
- Third-party relationships.
- Audience fit.
- Supply availability.
- Competitor timing.

Unlike the original's counterintuitive custom-console genre behavior, Studio Empire derives genre/audience fit from declared platform strategy, shipped catalog, marketing position, and audience behavior. It never makes the player's most-used genre secretly become the platform's worst fit.

23.6 Unit economics

```
unitMargin = wholesaleRevenuePerUnit - manufacturingCost - warrantyReserve
weeklyHardwareProfit = unitsSold * unitMargin - platformSupportCost
platformServiceRevenue = thirdPartySoftwareGross * platformFeeRate
```

A platform may sell hardware at a loss to build installed base, but this is an explicit choice with visible break-even projections.

23.7 Launch lineup

The player may coordinate owned games with the device launch. A launch title remains a normal game project with its own quality and schedule. Delaying the device for a title, launching without it, or rushing the title are distinct decisions.

23.8 Revisions and successors

- Revision: lowers cost, improves reliability, changes form factor, or adds bounded capability without resetting the ecosystem.
- Successor: new platform generation with migration, backward compatibility, and cannibalization decisions.
- Retirement: stops manufacturing but may retain network/support obligations.

The company may operate overlapping generations, but each consumes Hardware Operations Capacity.

23.9 Recalls and supply incidents

Severe undiscovered defects or risky suppliers may create incidents. Outcomes depend on real stored risk, not arbitrary punishment. Responses include repair program, replacement, limited recall, full recall, production pause, supplier switch, or denial. Each affects cost, trust, availability, and regulation risk.

---

24. Engine licensing, digital distribution, and ecosystem businesses

24.1 Engine licensing

Unlock requirements:

- Tier 5 or late Tier 4 groundwork.
- Mature versioned engine family.
- At least five shipped games on that family.
- SDK/toolchain project.
- Documentation maturity.
- Support capacity.
- R&D Director approval for full program.

License revenue depends on engine capability, reliability, documentation, market relevance, pricing, and support. Licensees generate support tickets and compatibility obligations. A licensed engine version cannot be silently altered.

24.2 Digital storefront

The storefront requires Digital Distribution and account infrastructure. It introduces:

- Catalog acquisition.
- Revenue share.
- Discovery policy.
- Regional operations abstraction.
- Fraud/refund/support load.
- Third-party developer relations.
- Platform trust.

The store affects reach and platform economics. It does not change a listed game's execution quality.

24.3 Cross-business dependencies

```
R&D creates technology and software-platform capability.
Hardware creates devices and physical platform capacity.
Production creates first-party games and IP.
Live Operations sustains persistent products and accounts.
Publishing/Store expands distribution and catalog.
```

No division should become an isolated money button. Each business creates obligations for at least one other division.

---

25. Industry-era content through 2050

The years after the present are fictional design space, not predictions. Content should use internally consistent invented names and avoid claiming real future products.

25.1 Era bands

Industry era	Core market/technology themes	Expected studio capability
1980-1989	Home computers, cartridges, basic 2D, simple audio, early console licensing	Garage and first-office foundation
1990-1999	16/32-bit transition, optical media, early 3D, richer audio, local multiplayer	First/Upgraded Office, team training
2000-2009	Broadband, HD, physics, online accounts, downloadable distribution	Upgraded Office and Technology Park path
2010-2019	Mobile scale, digital storefronts, live services, VR experiments, global online play	Technology Park, labs, Large/AAA
2020-2029	Cross-play, cloud services, advanced rendering, AI-assisted tooling, creator ecosystems	AAA, basic MMO, custom hardware
2030-2039	Cloud-native worlds, adaptive simulation, spatial platforms, advanced procedural production	Tier 5, directors, subscriptions, advanced live operations

Industry era	Core market/technology themes	Expected studio capability
2040-2050	Neural interfaces, persistent synthetic worlds, photonic/edge compute, deeply adaptive narrative, mature virtual economies	Full future campus and legacy projects

25.2 Future R&D branches

```
Advanced AI Tooling
├── Assisted Asset Pipeline
├── Adaptive QA
├── Synthetic Performance
└── Persistent Character Memory

Cloud-Native Simulation
├── Elastic World Shards
├── Seamless Device Handoff
├── Mass Participation Events
└── Persistent Synthetic Worlds

Spatial Computing
├── Advanced Presence
├── Full-Space Interaction
├── Haptic Environment
└── Neural Interface Safety

Future Engine Architecture
├── Neural Rendering
├── Real-Time World Synthesis
├── Autonomous Accessibility Adaptation
└── Continuous Engine Migration
```

25.3 Future Hardware branches

- Spatial headsets and room systems.
- Hybrid personal/cloud devices.
- Advanced haptic interfaces.
- Neural input/output prototypes.
- Edge-rendering nodes.
- Modular devices.
- Accessibility-first interfaces.
- Low-power photonic compute.

Each branch requires safety, reliability, manufacturing, price, audience fit, and developer ecosystem decisions. "Future" never means "strictly better in every dimension."

25.4 Legacy projects

Legacy projects are long, expensive, identity-defining undertakings available near the end of the Legacy campaign or in Endless Mode:

- A global game-development platform.
- A persistent world intended to operate for decades.
- A new hardware/interface category.
- An open engine and creator ecosystem.
- A preservation/archive initiative.
- A cross-generational franchise reboot.
- A studio university/research institute.

Legacy projects contribute to final score and change future market rules. They require directors and sustained divisional capacity.

26. Campaign completion and true Endless Mode

26.1 Scored completion

At the final configured campaign week:

1. Finish the current simulation-day settlement.
2. Enter `campaign_completion_pending`.
3. Freeze the score snapshot, not the studio.
4. Present results based on stored history.
5. Preserve active projects and services.
6. Offer End Campaign or Continue in Endless Mode.

The score snapshot includes:

- Solvency and profitability.
- Company valuation.
- Fans and reputation.
- Game catalog and legacy titles.
- Review performance without rewarding only one outlier.
- Employee development, retention, and leadership.
- Research and engine maturity.
- R&D and Hardware achievements.
- Hardware installed base and ecosystem health.
- Persistent-service lifetime health.
- Publishing/store/engine-license businesses.
- Awards and legacy projects.

26.2 Endless Mode principles

Endless Mode preserves everything and removes only the score deadline. It must continue producing meaningful decisions.

- All staff may still be hired, fired, promoted, retrained, placed on leave, or replaced.
- All projects, services, engines, platforms, labs, and offices remain active.
- The original campaign score is immutable.
- The original mode and completion date remain visible.
- New market and technology events continue.
- Rivals continue acting.
- Platforms continue entering, maturing, and retiring.
- Trends continue cycling.
- No arbitrary "all future research complete" flag is set.

26.3 Classic-mode continuation

A 35-year campaign scores around the 2030 industry band. Continuing advances through the remaining 2030-2050 authored future content. Those later actions do not alter the frozen Classic score but remain fully playable.

26.4 Legacy-mode continuation

A 50-year campaign scores after the 2050 authored band. Endless Mode then switches to procedural Frontier Cycles.

26.5 Procedural Frontier Cycles

After authored content ends, the simulation generates bounded future cycles:

- New platform generations every configured interval.
- New technology families assembled from compatible traits.
- Market shifts and genre revivals.
- Rival strategies and studio closures/acquisitions.
- New audience behavior.
- Cost and market growth using bounded curves.

Procedural content uses named traits and tradeoffs, not infinite "Graphics Level 987." A generation may improve compute while worsening cost, complexity, power, regulation, or accessibility.

26.6 Anti-inflation rules

- Use logarithmic display and bounded economic curves.
- Market population follows a saturating curve, not unbounded multiplication.
- Technology creates new capabilities and tradeoffs rather than permanent raw-score inflation.
- Employee stats use soft caps and mastery breadth.
- Review expectations use a bounded rolling market standard, not the player's all-time best game.
- Old games may revive through remasters, trends, ports, or adaptations without printing infinite money.
- Dormant entities use lower-frequency deterministic settlement without losing accounting accuracy.

26.7 Rolling expectations

Never repeat the original exploit where one exceptional title permanently raises an unreachable hidden target.

```
marketExpectation = boundedWeightedMedian(
  comparableRecentMarketTitles,
  studioRecentTitles,
  currentEraTechnicalBaseline
)
```

- Comparable titles are matched by size, platform era, genre class, and market tier.
- Old outliers decay out of the window.
- A great game raises reputation and future anticipation, but does not secretly doom every later project.
- Analyst Mode explains expectation changes.

---

27. Canonical domain model additions

The following sketches define ownership and relationships. Grok may adapt syntax to the existing codebase but may not collapse these into UI state.

27.1 Studio progression

```
type StudioTierId =
  | "garage"
  | "first_office"
  | "upgraded_office"
  | "technology_park"
  | "expanded_campus"

type StudioProgressionState = {
  currentTierId: StudioTierId
  enteredTierWeek: number
  technologyParkTenureWeeks: number
  completedTierIds: StudioTierId[]
  offers: Record<string, ProgressionOfferState>
  constructionProjectId?: string
}

type ProgressionOfferState = {
  offerId: string
  targetTierId: StudioTierId
  state: "hidden" | "discovered" | "eligible" | "offered" | "accepted" | "completed"
  discoveredWeek?: number
  offeredWeek?: number
  acceptedWeek?: number
  completedWeek?: number
  fixedCostSnapshot?: number
  requirementSnapshot?: RequirementEvaluation[]
}
```

27.2 Office definitions

```
type OfficeDefinition = {
  id: StudioTierId
  hqSeatCapacity: number
  productionSeatCapacity: number
  roleLockedSeats: Array<"rd_director" | "hardware_director">
  remoteRdCapacity: number
  remoteHardwareCapacity: number
  projectSlotDefinitions: ProjectSlotDefinition[]
  equipmentSlots: number
  weeklyOverhead: number
  moveOrUpgradeCost: number
  minimumRunwayWeeks: number
  capabilityUnlocks: CapabilityId[]
  eligibilityRules: EligibilityRule[]
}
```

27.3 HQ seats

```
type HqSeatState = {
  seatId: string
  officeId: StudioTierId
  seatType: "founder" | "production" | "rd_director" | "hardware_director"
  occupantId?: string
  locked: boolean
}
```

The founder seat cannot be vacated through firing. A director cannot occupy the wrong locked seat. Production seats cannot silently exceed office capacity.

27.4 Labs

```
type LabState = {
  id: "rd" | "hardware"
  lifecycle: "locked" | "proposed" | "constructing" | "active" | "paused"
  constructedWeek?: number
  actingLeadEmployeeId?: string
  directorEmployeeId?: string
  remoteStaffIds: string[]
  remoteCapacity: number
  weeklyBudget: number
  activeProjectId?: string
  backgroundThemeId?: string
  maturityByDomain: Record<string, number>
  documentationMaturity: number
}
```

27.5 Employees

```
type EmployeeState = {
  id: string
  employmentState: EmploymentState
  employeeClass: "hq_production" | "remote_rd" | "remote_hardware"
  hqSeatId?: string
  labId?: "rd" | "hardware"
  stats: {
    design: number
    technology: number
    speed: number
    research: number
  }
  disciplineMastery: Record<DisciplineId, number>
  genreMastery: Record<GenreId, number>
  leadership: LeadershipState
  certifications: string[]
  specializationId?: DisciplineId | "design_specialist" | "technology_specialist"
  assignmentIds: string[]
  energy: number
  fatigue: number
  morale: number
  burnoutRisk: number
  salaryWeekly: number
  joinedWeek: number
  departedWeek?: number
  trainingHistory: TrainingRecord[]
}
```

The founder retains a separate `FounderState` and must not be inserted into `EmployeeState` to reuse energy code.

27.6 Divisional leadership

```
type DirectorState = {
  employeeId: string
  division: "rd" | "hardware"
  appointedWeek: number
  portfolioPlanId?: string
  allocation: number
  handoffMaturity: number
  directReportIds: string[]
}
```

27.7 Live services

Use the `LiveServiceState` defined earlier and link it one-to-one with a released game build where applicable. Never infer a live service from title text, MMO genre labels, or a sales curve.

27.8 Hardware platforms

```
type OwnedPlatformState = {
  id: string
  lifecycle:
    | "thesis"
    | "architecture"
    | "prototype"
    | "developer_kit"
    | "manufacturing"
    | "launching"
    | "active"
    | "revision"
    | "retiring"
    | "retired"
  architectureVersionId: string
  formFactor: string
  targetAudience: AudienceId[]
  launchPrice: number
  manufacturingCost: number
  reliability: number
  installedBase: number
  activeUsers: number
  developerSupport: number
  firstPartyGameIds: string[]
  thirdPartyCatalogSize: number
  weeklySupportCost: number
  successorPlatformId?: string
}
```

28. Commands, events, and service ownership

28.1 Commands

All commands validate permissions, current state, capacity, cost, and idempotency key.

```
DISCOVER_PROGRESSION_OFFER
ACCEPT_OFFICE_OFFER
DECLINE_OFFICE_OFFER
START_OFFICE_CONSTRUCTION
PAUSE_OFFICE_CONSTRUCTION
CANCEL_OFFICE_CONSTRUCTION
COMPLETE_OFFICE_TRANSITION

START_CANDIDATE_SEARCH
MAKE_EMPLOYMENT_OFFER
WITHDRAW_EMPLOYMENT_OFFER
ASSIGN_EMPLOYEE
START_EMPLOYEE_TRAINING
START_RETRAINING
PROMOTE_EMPLOYEE
APPOINT_DIRECTOR
REMOVE_DIRECTOR
TERMINATE_EMPLOYEE
SEND_EMPLOYEE_ON_VACATION

PROPOSE_LAB
START_LAB_CONSTRUCTION
SET_LAB_BUDGET
ASSIGN_ACTING_LAB_LEAD
HIRE_REMOTE_LAB_EMPLOYEE
START_LAB_PROJECT
PAUSE_LAB_PROJECT
CANCEL_LAB_PROJECT
COMPLETE_LAB_PROJECT

START_AAA_GREENLIGHT
APPROVE_AAA_VERTICAL_SLICE
REVISE_AAA_SCOPE
APPROVE_MILESTONE
DELAY_MILESTONE
AUTHORIZE_OVERTIME

PLAN_LIVE_SERVICE
SET_SERVICE_CAPACITY
SET_MONETIZATION_MODEL
START_CONTENT_UPDATE
ANNOUNCE_SERVICE_SUNSET

COMPLETE_SERVICE_SUNSET

START_HARDWARE_THESIS
APPROVE_HARDWARE_ARCHITECTURE
BUILD_HARDWARE_PROTOTYPE
APPROVE_MANUFACTURING
SET_PLATFORM_LAUNCH
START_PLATFORM_REVISION
ANNOUNCE_PLATFORM_RETIREMENT

COMPLETE_CAMPAIGN_SCORE
ENTER_ENDLESS_MODE
```

28.2 Domain events

```
ProgressionOfferDiscovered
ProgressionOfferAccepted
OfficeConstructionStarted
OfficeTierChanged
HqSeatAdded
EmployeeHired
EmployeeOnboarded
EmployeeTrainingCompleted
EmployeeSpecialized
DirectorAppointed
EmployeeDeparted
LabConstructed
LabProjectStarted
LabProjectCompleted
TechnologyDiscovered
TechnologyResearched
PrototypeCompleted
EngineIntegrationRequested
AaaProjectGreenlit
AaaMilestoneResolved
LiveServiceLaunched
LiveServiceWeekSettled
LiveServiceSunset
OwnedPlatformLaunched
OwnedPlatformWeekSettled
OwnedPlatformRetired
CampaignScoreFrozen
EndlessModeEntered
```

28.3 Service ownership

System	Authority
Tier eligibility, offers, office transitions	`ProgressionService`
Seats, overhead, facility capacity	`OfficeService`
Candidate generation, hiring, firing, employment	`EmploymentService`
Energy, fatigue, morale, burnout, vacation	`EmployeeSimulationService`
Training, mastery, specialization, retraining	`TrainingService`
Assignments and work capacity	`AssignmentService`
R&D/Hardware lab lifecycle and budgets	`LabService`
Discovery, research, prototypes, maturity	`ResearchService`
Engine transfer and version creation	`EngineService`
AAA greenlight/milestones	`ProductionService` using the canonical project state machine
Online-service metrics and operations	`LiveOperationsService`
Owned platform lifecycle and unit economics	`HardwarePlatformService`
Era/platform/trend/rival events	`MarketService` and `TimelineService`
Immutable financial movement	`LedgerService`
Campaign score and Endless transition	`CampaignService`
Main-room character and cards	View-model selectors only

No service may mutate another service's state without a command/event transaction through the authoritative game store.

29. Project concurrency and assignment rules

29.1 Assignment is the real limit

Project slots indicate which kinds of simultaneous work are structurally possible. People and Standard Work Units determine whether work progresses.

29.2 Tier rules

Tier	Structural slots	Practical constraint
Garage	1 primary	Founder can do one primary action
First Office	1 main + tiny auxiliary	Four people total; Medium game often consumes most capacity
Upgraded Office	1 main + 1 auxiliary	Five people; auxiliary must have named assignments
Technology Park	1 main + 1 auxiliary + one per built lab + 1 live service	Acting lab leads reduce production capacity
Expanded Campus	1 main + 1 secondary + one per lab + up to 3 live services	Directors remove lead tax, but every activity still consumes people/operations capacity

29.3 Secondary production project

The Tier-5 secondary slot is intended for:

- Patch.
- Port.
- Remaster.
- Expansion.
- Prototype.
- Small game.
- Engine integration.
- Contract work.

Two simultaneous AAA projects are not supported by the initial eight-seat campus. That would require a later multi-team corporate expansion specification rather than pretending eight people can fully staff both.

29.4 Reassignment

- Reassignment is allowed at supported checkpoints.
- Mid-task switching incurs context loss based on documentation and task state.
- Removing the last qualified person from a required role blocks the next transition.
- The UI previews lost work, delay, and risk before confirmation.
- No assignment can total more than 100% of a person's capacity.

---

30. Economy and ledger requirements

30.1 Every movement is explicit

Additional ledger categories:

```
office_move
office_construction
office_overhead
hiring_search
signing_bonus
salary
```

```
severance
training
lab_construction
lab_budget
research_project
hardware_prototype
manufacturing_setup
hardware_unit_cost
hardware_unit_revenue
warranty_or_recall
subscription_revenue
subscription_refund
server_cost
live_operations
content_update
engine_license_revenue
engine_support_cost
storefront_revenue
storefront_partner_payment
```

30.2 Weekly settlement order

1. Existing debt obligations.
2. Salaries and benefits abstraction.
3. Office overhead.
4. Lab budgets.
5. Live-service infrastructure/support.
6. Hardware manufacturing/support.
7. Contractual partner obligations.
8. Game sales and service revenue.
9. Hardware/platform revenue.
10. Licensing/storefront revenue.
11. Tax/accounting abstraction if enabled.
12. Insolvency evaluation.

All entries are generated once, carry stable transaction IDs, and reconcile to cash.

30.3 Insolvency with ongoing services

When cash pressure occurs, the player receives staged options:

- Reduce lab budget.
- Pause a safe project.
- Cancel unspent construction.
- Renegotiate or cut marketing.
- Lay off staff with severance.
- Seek a publishing advance.
- Take a loan/bailout under existing finance rules.
- Sell or license IP.
- Sell a hardware/platform stake if that system exists.
- Announce service sunset.
- Move to a cheaper valid office only when continuity rules allow it.

The game cannot silently stop paying employees, turn off servers, or delete a platform to fix cash.

31. Player-facing flows

31.1 Office-offer screen

Display:

- Why the offer appeared.
- Requirements passed and failed.
- Current/new HQ seats.
- Current/new remote lab slots.
- Current/new project capability.
- New systems unlocked.
- One-time cost.
- New weekly overhead.
- Cash remaining.
- Estimated runway.
- Construction duration and disruption.
- Accept, Later, and View Details.

Do not market the move with a fake quality bonus.

31.2 Staff screen

Default mobile/vertical layout:

1. Compact company capacity header.
2. Filterable roster list.
3. One selected-person detail card.
4. Assignment, Train, Promote, Vacation, Compensation, and Terminate actions according to permission.
5. No simultaneous grid of eight full stat sheets.

31.3 Training screen

Show:

- Target person.
- Current stats/mastery/certifications.
- Available paths derived from tier and prerequisites.
- Time, RP, money, and lost-capacity cost.
- Diminishing-return/cooldown state.
- Result range.
- Required practical work before the next tier.

31.4 Division screens

R&D and Hardware each show:

- Director/acting lead.
- Remote team.
- Current project and milestone.
- Budget and burn.
- Capacity and risk.
- Backlog/roadmap.

- Recommendations.
- History.

No raw JSON and no fake stock-photo dashboard.

31.5 Live-service screen

Prioritize:

- Active users/subscribers.
- Capacity/reliability.
- Retention/churn.
- Content freshness.
- Community trust.
- Weekly revenue/cost/profit.
- Operations coverage.
- Current update/incident.
- Plan Update, Adjust Capacity, Change Offer, or Sunset.

Do not overload the room scene with these metrics.

31.6 Hardware-platform screen

Use one lifecycle-focused page:

- Current phase.
- Architecture and cost summary.
- Risk/issues.
- Installed base and supply after launch.
- Launch lineup/catalog.
- Unit economics.
- Developer support.
- Revision/successor/retirement action.

31.7 Notifications

Priority order:

1. Critical: bankruptcy, payroll failure, severe outage, recall, deadline/certification failure.
2. Important: office offer, director vacancy, lab completion, milestone decision, major platform event.
3. Informational: employee level, research discovery, normal sales milestone.
4. Background: routine weekly settlements and minor market movement.

Only one blocking notification at a time. Background events go to history. Deduplicate repeated warnings and use cooldowns.

---

32. Configuration sketch

All values must live in authored configuration and be schema-validated.

```
{
  "campaignModes": {
    "classic_35": { "campaignYears": 35, "industryStart": 1980, "industryEnd": 2030 },
    "legacy_50": { "campaignYears": 50, "industryStart": 1980, "industryEnd": 2050 }
  },
  "officeProgression": {
    "garage": {
      "hqSeats": 1,
```

```
        "productionSeats": 1,
        "weeklyOverhead": 0
      },
      "first_office": {
        "hqSeats": 4,
        "productionSeats": 4,
        "cashGate": 1000000,
        "moveCost": 150000,
        "minimumRunwayWeeks": 26,
        "minimumReleases": 5,
        "minimumFans": 1000
      },
      "upgraded_office": {
        "hqSeats": 5,
        "productionSeats": 5,
        "cashGate": 5000000,
        "upgradeCost": 500000,
        "minimumRunwayWeeks": 39,
        "minimumTenureWeeks": 52
      },
      "technology_park": {
        "hqSeats": 6,
        "productionSeats": 6,
        "cashGate": 16000000,
        "moveCost": 8000000,
        "minimumRunwayWeeks": 52,
        "minimumTenureWeeks": 104,
        "remoteRdCapacity": 2,
        "remoteHardwareCapacity": 2
      },
      "expanded_campus": {
        "hqSeats": 8,
        "productionSeats": 6,
        "roleLockedSeats": ["rd_director", "hardware_director"],

        "cashGate": 100000000,
        "expansionCost": 50000000,
        "minimumRunwayWeeks": 104,
        "technologyParkTenureWeeks": 624,
        "remoteRdCapacity": 4,
        "remoteHardwareCapacity": 4
      }
    },
    "rp": {
      "founderPerDayDeveloping": 0.15,
      "founderPerDayResearching": 0.25,
      "founderPerDayReport": 0.2,
      "founderPerDayContract": 0.1,
      "marketWeeklyBase": 0.4,
      "staffPerWeekByTier": {
        "first_office": 1,
        "upgraded_office": 2,
        "technology_park": 4,
        "expanded_campus": 4
      }
    },
    "labs": {
      "designSpecialist": { "design": 700, "research": 400 },
      "technologySpecialist": { "technology": 700, "research": 400 },
      "directorLeadershipCertification": 2
    },
    "expandedCampus": {
      "offerAfterTechnologyParkYears": 12,
      "graphicsLevelGate": null
    }
  }
```

The `graphicsLevelGate: null` entry is deliberate documentation against accidental reintroduction.

---

33. Save, load, and migration

33.1 Persist at minimum

- Campaign mode, progress, industry era, score state, and Endless state.
- Current studio tier and tenure counters.
- Progression offers and construction state.

- HQ seats and occupants.
- Founder state.
- Every current and departed employee.
- Training, specialization, leadership, energy, morale, and assignment state.
- Labs, directors, acting leads, remote staff, budgets, maturity, and projects.
- AAA milestone state.
- Live-service state and settlement cursor.
- Owned-platform lifecycle and unit history.
- All associated ledger entries.
- Deterministic event cursors and idempotency keys.

33.2 Migration from the existing Garage build

For a garage save:

- Set `currentTierId = "garage"`.
- Create only the founder HQ seat.
- Preserve founder stats/mastery without employee energy.
- Set all later capacities to zero.
- Evaluate first-office eligibility from stored releases, fans, cash flow, and campaign time.
- Do not auto-move even if the save qualifies.
- Queue the offer once after load if newly eligible.

33.3 Migration from any experimental employee build

- Resolve every person to founder versus hired employee.
- Never convert founder energy into employee energy.
- Map existing office labels to a tier only when capacity and progression history support it.
- If headcount exceeds the mapped capacity, enter a non-destructive `migration_capacity_review` state. Do not delete staff.
- Preserve money and ledger history; add an explicit migration adjustment only when unavoidable.
- Preserve released-game snapshots exactly.

33.4 Determinism tests

Saving and loading immediately before any of the following must produce the same result:

- Office-offer discovery.
- Candidate generation.
- Hire acceptance.
- Training completion.
- Lab-project milestone.
- AAA milestone review.
- Live-service weekly settlement.
- Hardware prototype result.
- Platform launch week.
- Campaign scoring.

---

34. Balance harness

34.1 Required simulated player profiles

1. Competent conservative: delays moves until cash is safe.
2. Aggressive expander: accepts each move at the first legal moment.
3. Overhirer: fills seats immediately with expensive staff.
4. Underhirer: attempts Medium/Large work with too few people.
5. Training-first: spends heavily on staff and delays releases.
6. Publisher-dependent: uses deals to build reach.
7. Self-publisher: accepts lower reach and keeps rights/revenue.
8. Lab-heavy: invests in R&D/Hardware at the expense of releases.
9. AAA-heavy: runs long productions and large support obligations.
10. Live-service overextension: launches more operations than capacity.
11. Hardware gamble: invests early in a platform with limited cash.
12. Late bloomer: reaches Technology Park near the score deadline.
13. Garage forever: deliberately refuses every move.
14. Roster reset: fires/retrains most staff in Endless Mode.

34.2 Target outcomes

- A competent player reaches the first-office offer after roughly 5-8 releases.
- Moving immediately at exactly $1M is possible but financially tense.
- Filling Tier 2 increases potential and burn; it does not instantly produce perfect games.
- Tier 3 meaningfully improves training and specialization before the Technology Park.
- The median Legacy run reaches Technology Park early enough for the 12-year expansion to appear before scoring.
- The median Classic run may see the expansion late or in Endless Mode; this is acceptable.
- A well-run Tier-4 studio can build both labs but feels the acting-lead tradeoff.
- Tier 5 visibly frees the production group while making directors necessary for advanced projects.
- An MMO can be very profitable, but neglect eventually produces churn and losses.
- A custom platform can succeed, fail, or break even for understandable reasons.
- Endless Mode remains stable for at least 100 additional campaign years under automated soak tests.

34.3 Assertions

- No required milestone has a success rate below the configured accessibility target for a competent strategy.
- No single title, platform, MMO, subscription, engine license, or lab idle mode prints unbounded money or RP.
- No office move guarantees solvency.
- No office refusal triggers game over by itself.
- No campaign mode hides a required core progression capability permanently; Endless preserves it.

---

35. Publishing, contracts, sequels, franchises, and marketing across progression

These systems are part of the campaign spine. They must scale with the studio rather than disappear when offices unlock.

35.1 Contracts

Tier	Contract role
Garage	Emergency income and early mastery; occupies founder's primary assignment
First Office	Small team contracts; teaches assignments and deadlines
Upgraded Office	Larger technical/design contracts; auxiliary slot may keep the main company project moving
Technology Park	Specialist consulting, engine work, ports, platform commissions
Expanded Campus	Major co-development, platform partnerships, research/hardware contracts

Every contract has requirements, deadline, work, payment, penalty, and deterministic resolution. It is never instant cash.

35.2 Publishing-deal board

Retain the agreed early model:

- Initial board size: 3 deals.
- Natural refresh every season: 84 simulation days.
- One paid manual refresh per season.
- Initial garage refresh cost: $2,000, scaled by tier/economy.
- Manual refresh creates a new deterministic board and marks the season refresh used.
- Backing out of the board does not reroll it.

Late-garage discovery remains valid after at least one release or 500 fans, if that feature is implemented in the Garage slice. Tier 2 makes publishing central for Medium games.

35.3 Deal fields

```
type PublishingDeal = {
  id: string
  publisherId: string
  requiredTopicId?: string
  requiredGenreId?: string
  allowedPlatformIds?: string[]
  requiredSize?: "small" | "medium" | "large" | "aaa"
  minimumDisplayedReview: number
  deadlineWeek?: number
  advance: number
  royaltyRate: number
  marketingReach: number
  failurePenalty: number
  rights: {
    sequelOption: boolean
    platformExclusivity: boolean
    ipOwnershipShare: number
    subscriptionInclusionOption: boolean
  }
  offeredWeek: number
  expiresWeek: number
}
```

Rights become more important as the company grows. A large advance with sequel rights is a real long-term tradeoff.

35.4 Publishing negotiation by tier

- Garage: fixed offers only.
- First Office: choose among fixed offers; paid board refresh.
- Upgraded Office: negotiate one economic term at the risk of rejection.
- Technology Park: negotiate advance, royalty, marketing, platform exclusivity, and sequel option within publisher tolerance.
- Expanded Campus: act as developer, self-publisher, co-publisher, or publisher of third-party titles.

Negotiation uses reputation, prior delivery, relationship, market leverage, and deal risk. Reloading cannot reroll an accepted/rejected counteroffer.

35.5 Sequels

Retain the established sequel rules:

- Unlock after two releases or the Series Continuity research path.
- Source game must be released or dormant, never cancelled.
- Full timing benefit begins after roughly 40 weeks.
- A sequel on a newer compatible engine receives stronger technical freshness than one on the same engine.
- Source-game fans increase awareness, not execution quality.
- Premature sequels risk fatigue and lower novelty.
- The player still chooses platform, size, engine, production intent, features, and staff.

Initial quality-side timing modifiers remain a starting balance reference:

```
properGapNewEngine = 1.12
properGapSameEngine = 1.05
early = 0.97
premature = 0.88
```

Do not double-punish the same repetition through both sequel timing and the generic repetition model. The scoring pipeline must declare which factor owns each penalty.

35.6 Franchise state

```
type FranchiseState = {
  id: string
  gameIds: string[]
  awareness: number
  reputation: number
  fatigue: number
  dormantWeeks: number
  audienceExpectation: number
  rightsByPartner: Record<string, number>
}
```

Good sequels build franchise reputation. Repeated low-effort releases, broken services, and exploitative monetization build fatigue. Time, reinvention, a strong reboot, or a different format can reduce fatigue.

35.7 Ports, remasters, remakes, and expansions

Type	Definition
Port	Existing game adapted to another platform build; preserves core quality, adds technical/certification work
Remaster	Existing game with improved presentation/compatibility; moderate work and expectation
Remake	New project based on an old title; uses full production pipeline
Expansion	Substantial content attached to an existing eligible game/service

These store explicit source links. They are not sequels with renamed labels.

35.8 Marketing progression

Tier	Available marketing class
Garage	Word of mouth, basic press/contact when discovered
First Office	Small campaigns and publisher reach
Upgraded Office	Targeted campaigns, interviews, convention booths
Technology Park	Global campaign, major convention, AAA launch support

Tier	Available marketing class
Expanded Campus	Portfolio campaign, platform launch, subscription/catalog campaign, owned convention

Marketing settlement feeds awareness, reach, wishlists/preorders if modeled, and launch momentum. It never feeds the execution-quality calculation.

35.9 Employee roles inside production

Every Medium-or-larger project identifies real responsibilities:

- Project Lead.
- Stage/milestone Lead.
- Discipline owners for selected high-intent areas.
- QA owner for Large/AAA.
- Platform-build owner for multi-platform work.
- Live Operations owner for persistent products.

One person may hold several responsibilities on a small team, but responsibility overload reduces attention and estimation accuracy. A title cannot have nine perfect "owners" with only three people.

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36. Acceptance tests

The following are minimum behavioral tests. UI snapshot tests do not replace domain assertions.

36.1 Campaign and timeline

1. Campaign creation accepts only enabled modes `classic_35` and `legacy_50`.
2. Classic has 35 campaign years and an authored industry target near 2030.
3. Legacy has 50 campaign years and an authored industry target of 2050.
4. Campaign time and industry era persist independently.
5. Identical seeds and commands produce identical era events.
6. First-office tutorial windows are bounded and do not stretch proportionally across the whole Legacy timeline.
7. Date/era alone never grants an office tier.
8. A completed campaign freezes its score snapshot.
9. Endless continuation does not alter the frozen score.
10. Classic Endless can continue through authored 2030-2050 content.
11. Legacy Endless enters procedural Frontier Cycles after authored content.

36.2 Garage to First Office

12. Garage capacity is exactly one founder HQ seat.
13. Hiring is unreachable in the garage.
14. First-office eligibility evaluates releases, fans, profit/cash flow, cash, time, and runway.
15. Reaching $1M without the non-financial proof does not complete the move.
16. Meeting non-financial proof without affordability may reveal but cannot accept the offer.
17. The offer is deterministic and persists after decline.
18. Reopening it does not change cost or requirements.
19. Acceptance reserves funds once.
20. Move completion preserves all games, engines, research, mastery, money history, and market state.
21. First Office has exactly four total HQ seats, including founder.

22. The move adds no free employee.

36.3 Hiring and First Office

23. First Office permits at most three hired HQ employees.
24. Candidate generation is deterministic for search identity and seed.
25. Reopening a candidate slate does not reroll it.
26. A new paid search creates a new stable slate.
27. An accepted hire enters onboarding.
28. Welcome Training reduces onboarding/ramp cost.
29. New-hire coordination loss affects work/bugs, not concept fit.
30. Salary and signing costs reconcile to the ledger.
31. A full office blocks a fourth hired employee.
32. The founder is never counted in employee-energy simulation.
33. Tier-2 weekly staff RP counts only qualifying hired production employees.
34. A full Tier-2 staff week produces at most 3 base weekly staff RP.
35. Vacation/onboarding participation prorates or removes RP according to config.

36.4 First Office to Upgraded Office

36. Upgrade eligibility requires all four seats filled.
37. It requires four releases since the move.
38. It requires cohesion and Management Fundamentals.
39. It requires at least 52 weeks of Tier-2 tenure.
40. It validates cash and post-upgrade runway.
41. Upgrade completion preserves the same office identity/background family.
42. Upgraded Office has exactly five HQ seats.
43. It adds no free employee.
44. Training Center capabilities remain unavailable before completion.
45. Tier-3 weekly staff RP counts at most four hired production employees.
46. A full Tier-3 staff week produces at most 8 base weekly staff RP.

36.5 Training and staff lifecycle

47. Training consumes calendar time and person capacity.
48. Training cannot settle twice after save/load.
49. Repeated training invokes configured diminishing returns/cooldown.
50. Discipline specialization checks its actual prerequisites.
51. Specialization improves relevant fit and does not multiply all work.
52. Retraining removes the old active bonus during transition.
53. Retraining preserves the employee's identity and transferable history.
54. Firing writes severance and departure events.
55. Firing removes personal future contribution but not documented studio knowledge.
56. Departed employee IDs are not reused.
57. Burnout can reduce work and cause leave/departure.
58. The founder never acquires hired-employee energy through training or migration.

36.6 Technology Park

59. Technology Park eligibility requires five occupied Tier-3 HQ seats.
60. It checks total releases, profitable scaled games, specialization, operating profit, tenure, cash, and runway.
61. Acceptance cannot occur inside an atomic release settlement.
62. Completion charges exactly one move cost.
63. Technology Park has exactly six HQ seats.
64. It adds one vacant normal production seat.
65. It starts `technologyParkTenureWeeks` at zero upon completion.
66. Pausing the game does not advance campus tenure.
67. Saving/loading does not reset campus tenure.
68. A full Tier-4 production staff week produces at most 20 base weekly staff RP.

36.7 R&D Lab

69. R&D proposal is hidden before Technology Park ownership.
70. R&D construction requires a valid Design Specialist.
71. Design Specialist training checks Design, Research, experience, RP, money, and time.
72. R&D construction completion creates no free remote employee.
73. Tier 4 permits at most two remote R&D employees.
74. A Tier-4 R&D major project requires an acting lead.
75. Acting-lead allocation reduces that HQ person's production availability.
76. Research discovery does not modify an engine.
77. Research completion does not modify an engine.
78. Prototype completion does not silently modify active games.
79. Engine transfer creates a separate integration project and new version.
80. Lab idle research is capped, costs money, and cannot print unbounded RP.

36.8 Hardware Lab

81. Hardware proposal is hidden before R&D and Hardware Architecture research.
82. Hardware construction requires a valid Technology Specialist.
83. Tier 4 permits at most two remote Hardware employees.
84. Hardware major projects require an acting lead before Tier 5.
85. Running both labs may reduce available production leadership twice.
86. A custom platform cannot skip thesis, architecture, prototype, manufacturing, and launch.
87. Platform unit economics reconcile to the ledger.
88. Reliability investment affects stored risk rather than random punishment.
89. A revision preserves platform ecosystem identity.
90. Retirement preserves sales/support history.

36.9 AAA

91. AAA discovery requires Large-game proof and an R&D Lab.
92. AAA remains unavailable until the R&D major project completes.
93. AAA uses the canonical game project plus additional milestone states.
94. Greenlight validates engine, staff, budget, capacity, platforms, and support plan.
95. Vertical-slice findings derive from stored risks and work.
96. Milestone decisions consume money/time/capacity as declared.

97. Overtime adds bounded output and increases fatigue/defect/retention risk.
98. Marketing support changes awareness, not execution quality.
99. Hardware optimization changes relevant performance/certification factors, not arbitrary reviews.
100. AAA release creates support obligations.

36.10 MMO/live services

101. MMO is modeled independently from genre.
102. MMORPG requires MMO capability and RPG/persistent-world requirements.
103. A live service has an explicit lifecycle linked to a released game.
104. Weekly settlement order is stable and idempotent.
105. Server capacity shortfall can cause reliability/onboarding/churn problems.
106. Content freshness decays without support.
107. Operations undercoverage has visible consequences.
108. Service revenue subtracts infrastructure, staffing, support, and content cost.
109. Expansion effects depend on quality and service state; they do not blindly reset launch sales.
110. Tier 4 supports only one persistent-service operation slot by default.
111. A service may be placed into maintenance or sunset.
112. Sunset stops future billing according to policy and preserves history.

36.11 Expanded Campus and directors

113. Expanded Campus discovery occurs after 624 Technology Park tenure weeks.
114. No graphics/GPU level is evaluated for the offer.
115. Cash and runway are still required for acceptance.
116. Completion creates exactly eight HQ seats.
117. Six seats belong to the production group including founder.
118. The seventh seat accepts only an R&D Director.
119. The eighth seat accepts only a Hardware Director.
120. Completion does not create either director automatically.
121. Remote capacity remains effectively constrained while a director seat is vacant.
122. A qualified existing employee may be promoted while preserving identity/history.
123. Promoting a production employee opens that production seat.
124. R&D Director vacancy prevents new Tier-5 R&D major projects.
125. Hardware Director vacancy prevents new Tier-5 Hardware major projects.
126. Active projects pause safely or use a valid acting lead after director loss.
127. Tier-5 general staff RP still counts only five hired production employees.

36.12 Subscriptions, stores, and engines

128. Subscription is a monetization model, not a genre or quality modifier.
129. Subscriber revenue is calculated from real subscribers and price.
130. Refunds, fees, infrastructure, support, and content costs are accounted.
131. Aggressive monetization may affect conversion/trust/churn but not rewrite production truth.
132. Catalog subscription requires Tier 5, account/distribution capability, and sufficient content.
133. Engine licensing requires a versioned engine, SDK, documentation, and support capacity.
134. Licensees create support obligations.
135. Storefront revenue and partner payments remain separate ledger channels.

36.13 Vertical UI

136. Main scene renders no more than one full-size focus character.
137. Off-screen employees continue producing work normally.
138. Focus selection alone never reassigns work.
139. Remote roster chips show real assignment/status data.
140. Locked destinations cannot be opened through URL, keyboard, or stale save route.
141. The room does not own formulas or simulation state.
142. Only one blocking notification is displayed at a time.
143. Routine weekly events go to history rather than popup spam.

36.14 Endless stability

144. Every employee action remains available after campaign scoring.
145. New platforms and technologies continue after score.
146. Procedural Frontier content uses bounded traits rather than infinite raw levels.
147. Market growth saturates and does not overflow.
148. Rolling expectations decay old outliers.
149. A perfect old title cannot permanently doom later reviews.
150. A 100-year post-score soak test remains deterministic and financially numerically stable.

---

37. Required implementation order for Grok

Do not build the entire late game in one uncontrolled pass. Implement in dependency order and stop at each checkpoint for review.

Checkpoint 0: Protect the existing Garage

- Run all current Garage tests.
- Record save schema and migration fixtures.
- Confirm founder remains separate and has no employee energy.
- Confirm no current quality/review/sales formula moves into UI.
- Add feature flags for every later system.

Stop condition: Existing Garage path behaves identically before new progression is enabled.

Checkpoint 1: Progression and office foundation

- Add campaign modes and industry-era mapping.
- Add `StudioProgressionState`.
- Add data-driven office definitions.
- Add HQ seat model.
- Add offer state machine.
- Add move/renovation transaction and ledger handling.
- Implement Garage -> First Office only.
- Add save migration.

Stop condition: A real Garage save can earn, defer, accept, save during, and complete the first-office transition without employees yet being fake-created.

Checkpoint 2: First Office employees

- Candidate search.
- Hiring/offers.
- Three employee maximum.
- Onboarding/cohesion.
- Energy, fatigue, morale, vacation.
- Payroll.
- Assignments.
- Welcome/basic training.
- Tier-2 RP.
- Vertical remote roster presentation.

Stop condition: Player can run a four-person total company and make Medium games with real staffing consequences.

Checkpoint 3: Upgraded Office and Training Center

- Tier-2 proof and renovation offer.
- Fifth seat.
- Advanced training.
- Mentorship.
- Leadership state.
- Discipline specialization/retraining.
- Auxiliary project capacity.
- Large-game publishing bridge.
- Tier-3 RP.

Stop condition: The same office visibly and mechanically upgrades, and a five-person team can deliberately prepare for scale.

Checkpoint 4: Technology Park and labs

- Tier-3 proof and Technology Park move.
- Sixth seat.
- Technology Park tenure counter.
- Design/Technology Specialist paths.
- R&D construction, staff, budgets, projects.
- Hardware dependency, construction, staff, projects.
- Acting-lead capacity tradeoff.
- Remote lab roster.
- Tier-4 RP.

Stop condition: Both labs can run, pause, survive save/load, and visibly compete with production for leadership.

Checkpoint 5: AAA

- Large-game proof.
- AAA R&D project.
- Greenlight package.
- Vertical slice.
- Milestones, alpha, beta, certification.
- Support obligations.

- Lab-support projects.

Stop condition: One complete AAA game can be developed, cancelled at valid checkpoints, released, supported, and audited.

Checkpoint 6: Persistent online games

- Network research tree.
- Service lifecycle.
- Capacity/reliability.
- Active users, retention, churn.
- Operations Capacity.
- Updates/expansions.
- Tier-4 one-service limit.
- Sunset.

Stop condition: One MMO can succeed or fail for explainable stored reasons and cannot generate passive profit without operations.

Checkpoint 7: Expanded Campus and directors

- 12-year offer.
- Construction state.
- Eight seats with role locks.
- Director eligibility, promotion, external search, vacancy, firing, handoff.
- Expanded lab capacity.
- Production group preserved at six.
- Additional service/secondary capacity.

Stop condition: Campus expansion removes acting-lead tax only after valid directors are appointed.

Checkpoint 8: Advanced endgame businesses

- Subscription models.
- MMORPG framework.
- Multi-service portfolio.
- Engine licensing.
- Digital storefront/catalog subscription.
- Full custom-hardware lifecycle.
- Director roadmaps.

Stop condition: Each business has costs, staffing, state, failure, and shutdown paths; none is a passive money button.

Checkpoint 9: 2030-2050 and Endless

- Future-era authored data.
- Classic continuation through future content.
- Legacy scoring at 2050.
- Procedural Frontier Cycles.
- Rolling expectations.
- Saturating economy/market curves.
- 100-year soak harness.

Stop condition: Market, rivals, platforms, technology, staff, and finances continue coherently for 100 post-score years.

Checkpoint 10: Final vertical UX pass

- Replace temporary screens only after systems are stable.
- Preserve one-character main scene.
- Validate all destinations, cards, responsive layouts, focus management, and notification priority.
- No formula migration into components.

Stop condition: Complete campaign can be played vertically without debug dashboards or inaccessible decisions.

---

38. Required Grok build report at every checkpoint

Grok must provide:

1. Exact scope completed.
2. Files added/changed.
3. State/schema changes.
4. Commands/events added.
5. Config values added.
6. Save migration behavior.
7. Automated tests and result counts.
8. Manual player path tested.
9. Screenshots for every changed player-facing state.
10. Known limitations.
11. Explicit confirmation that authoritative calculations remain outside UI.
12. Explicit confirmation that the founder remains outside employee-energy logic.
13. Explicit confirmation that no future checkpoint was faked with empty navigation.
14. A stop for Del's review before proceeding.

Grok may improve architecture when highly confident, but it must explain any deviation from this specification and prove that the player-facing rule remains equivalent or better. It may not silently change seat counts, office gates, campaign modes, the 12-year expansion, director roles, or the one-character scene.

---

39. Anti-drift rules

1. Do not rebuild the Garage loop while adding offices.
2. Do not use UI-local booleans as progression authority.
3. Do not call the founder an employee to reuse code.
4. Do not render more than one full-size character in the vertical home scene.
5. Do not change 1 -> 4 -> 5 -> 6 -> 8 total HQ seats.
6. Do not let the final two seats behave as normal developers.
7. Do not require Level-7 graphics/GPU technology for the campus expansion.
8. Do not unlock Tier 5 before 12 full Technology Park years.
9. Do not grant labs merely on office entry.
10. Do not grant Hardware before R&D, Hardware research, and a Technology Specialist.
11. Do not grant AAA merely on office entry.
12. Do not model MMO as a genre.
13. Do not model subscriptions as automatic profit.
14. Do not let marketing change review quality.

15. Do not let opening screens create work, RP, sales, candidates, or events.
16. Do not freeze the market after the score screen.
17. Do not use the player's all-time best title as an eternal hidden review target.
18. Do not erase employees, games, engines, labs, or ledger history during office changes.
19. Do not add empty R&D, Hardware, AAA, MMO, or future-tech buttons before their loop works.
20. Do not proceed past a checkpoint without the requested report and review stop.

---

40. Final completion definition

The full progression is complete only when a player can:

1. Begin alone in the existing Garage.
2. Earn and optionally defer the first-office offer.
3. Move into a four-seat total office and hire exactly three employees.
4. Onboard, assign, train, pay, rest, retain, fire, and replace those employees.
5. Earn an in-place upgrade to a five-seat office with a meaningful Training Center.
6. Train specialists, use mentorship, retrain careers, and prepare Large games.
7. Earn and afford the six-seat Technology Park.
8. Qualify a Design Specialist and build a functional R&D Lab.
9. Research Hardware Architecture, qualify a Technology Specialist, and build a functional Hardware Lab.
10. Feel the acting-lead tradeoff while the six-person production group shares leadership with both labs.
11. Prove Large-game mastery and research AAA.
12. Develop and support AAA games through real milestone states.
13. Research and launch a persistent online game/MMO with real operations, capacity, churn, updates, and shutdown.
14. Receive the campus expansion offer after 12 exact Technology Park years.
15. Expand to eight HQ seats without a graphics-level gate.
16. Appoint dedicated R&D and Hardware directors into role-locked seats.
17. Keep the six-person production group intact while directors manage expanded remote divisions.
18. Develop MMORPGs, subscriptions, engine licensing, digital distribution, and custom platforms through real systems.
19. Reach future-era 2030-2050 content in the Legacy campaign or through Classic Endless continuation.
20. Complete the scored campaign without losing active work.
21. Continue indefinitely with changing platforms, technologies, markets, rivals, services, hardware, and staff.
22. Fire, hire, promote, or retrain the entire postgame organization if desired.
23. Save and load at every major boundary with deterministic results.
24. Understand why each success, failure, delay, cost, review, sale, churn event, and progression offer occurred.

That is the intended fantasy: one person at a garage desk becomes the leader of a studio, then the leader of specialized divisions, then the steward of a decades-long creative and technology institution-without ever losing the intimate, readable "one person working in the room" presentation that makes the vertical version possible.

Source notes

Research references used to verify the original baseline:

- Greenheart Games: official Game Dev Tycoon overview and FAQ

- Steam Community: Office Requirements
- GameFAQs: Training Guide
- Argade: AAA unlock requirements
- Greenheart Games Forum: campaign-length and continuation behavior

Community sources can disagree across game versions. They are used here to establish the original's dependency shape and documented reference values. Studio Empire's locked rules are the intentional design defined above, not a promise of exact version parity with Game Dev Tycoon.